16、Dcm\Dem\CanTp\Nvm\Crc调通Dcm响应

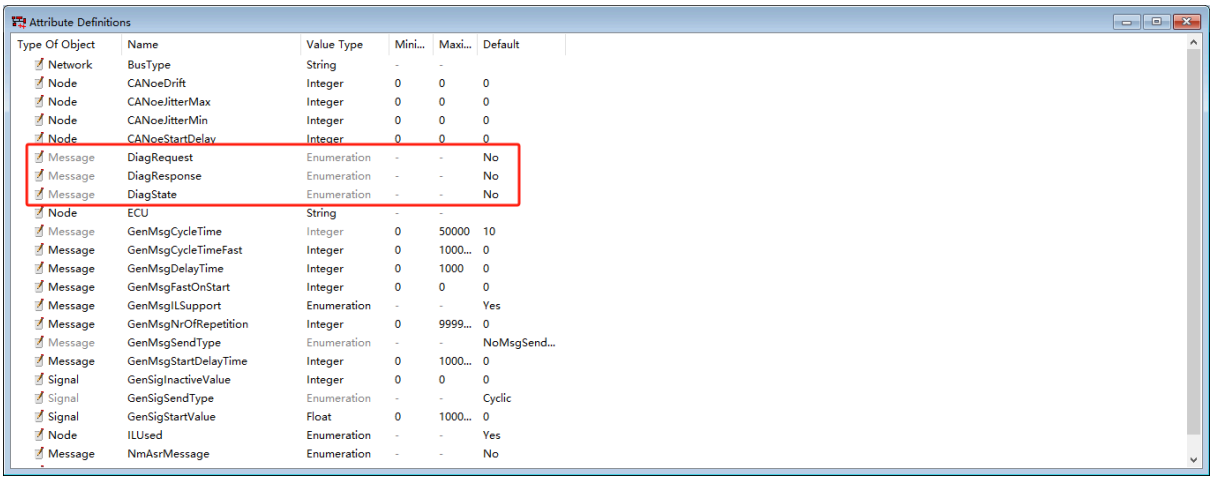
卖家'漩涡遨游aa' 专属资料，非在漩涡遨游aa 购买资料，不提供答疑(一次购买仅提供有限次数10次答疑)

集成Dcm诊断相关联的模块Dcm\Dem\Nvm\Crc\Ea，由于Sip包没有FEE，所以只能用的Ea，但是我们不生成Ea，本章节主要以配置Dcm\Dem为主，搭好环境调通诊断通信基础功能(注意：后续是移植了FEE)

一、修改DBC，添加物理寻址的报文（修改不了dbc直接用我提供的）

dbc修改需要安装vetor canoe工具

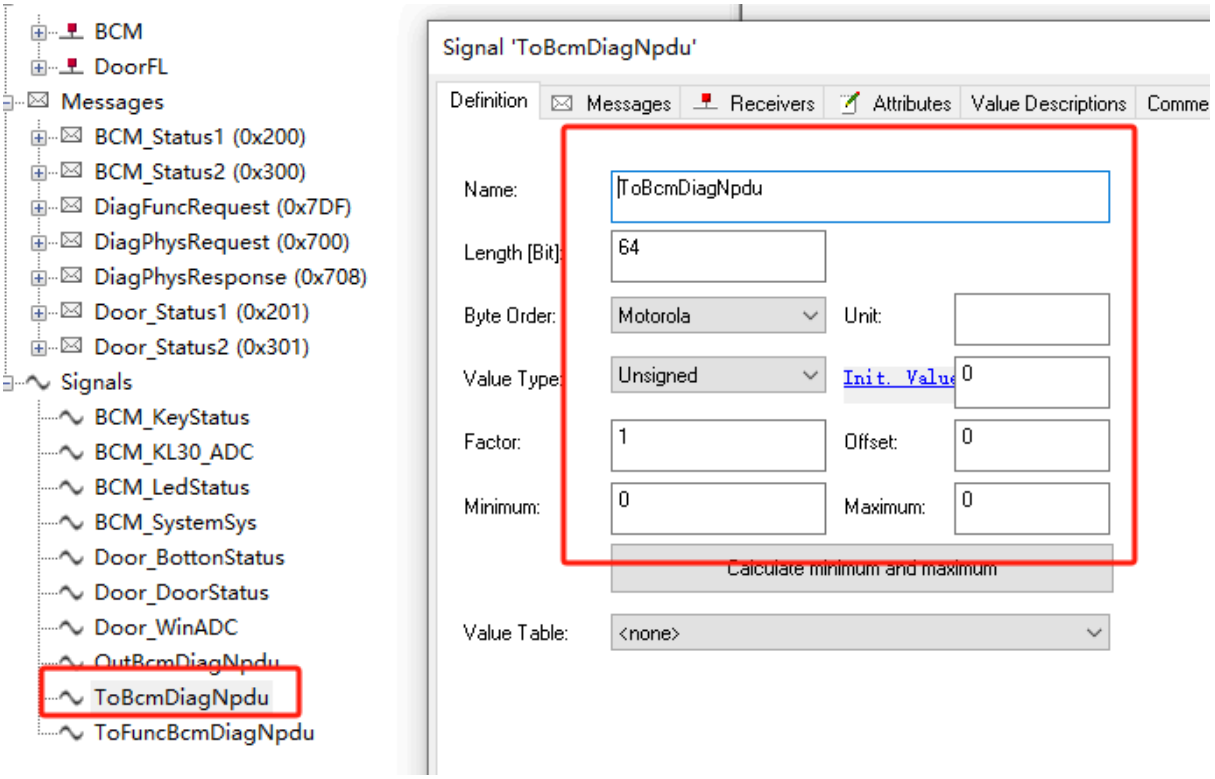
添加属性



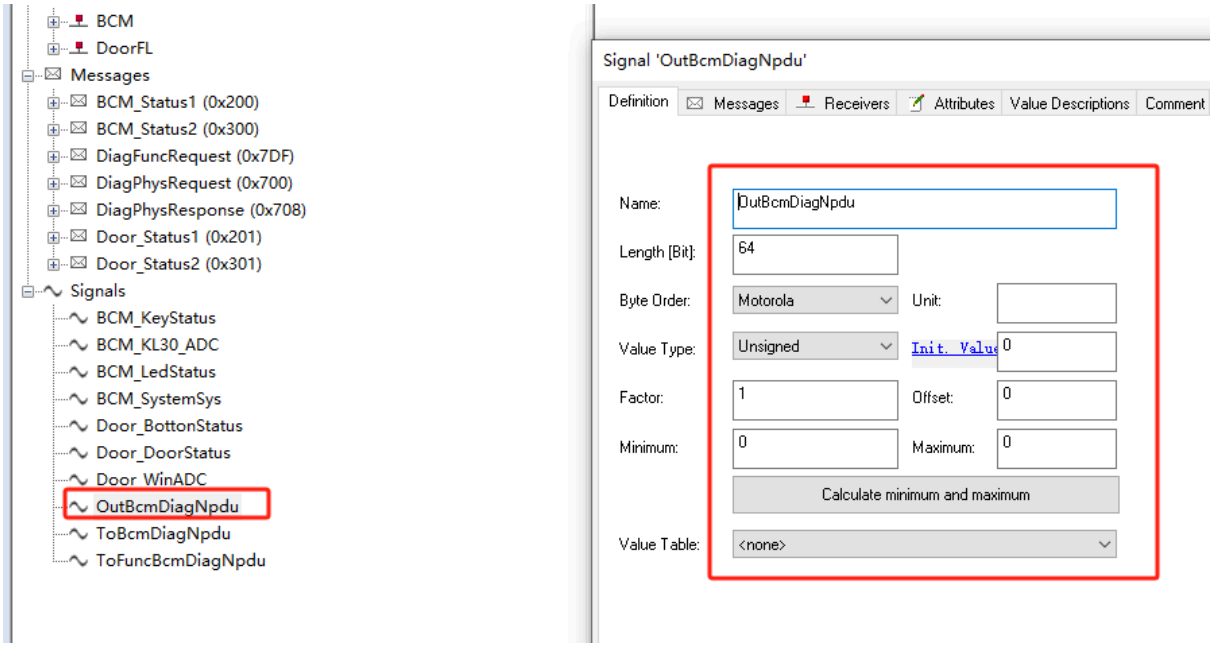
Attribute Definitions

Type Of Object	Name	Value Type	Mini...	Maxi...	Default
Network	BusType	String	-	-	
Node	CANoeDrift	Integer	0	0	0
Node	CANoeJitterMax	Integer	0	0	0
Node	CANoeJitterMin	Integer	0	0	0
Node	CANoeStartDelay	Integer	0	0	0
Message	DiagRequest	Enumeration	-	-	No
Message	DiagResponse	Enumeration	-	-	No
Message	DiagState	Enumeration	-	-	No
Node	ECU	String	-	-	
Message	GenMsgCycleTime	Integer	0	50000	10
Message	GenMsgCycleTimeFast	Integer	0	1000...	0
Message	GenMsgDelayTime	Integer	0	1000	0
Message	GenMsgFastOnStart	Integer	0	0	0
Message	GenMsgILSupport	Enumeration	-	-	Yes
Message	GenMsgNrOfRepetition	Integer	0	9999...	0
Message	GenMsgSendType	Enumeration	-	-	NoMsgSend...
Message	GenMsgStartDelayTime	Integer	0	1000...	0
Signal	GenSigInactiveValue	Integer	0	0	0
Signal	GenSigSendType	Enumeration	-	-	Cyclic
Signal	GenSigStartValue	Float	0	1000...	0
Node	ILUsed	Enumeration	-	-	Yes
Message	NmAsrMessage	Enumeration	-	-	No

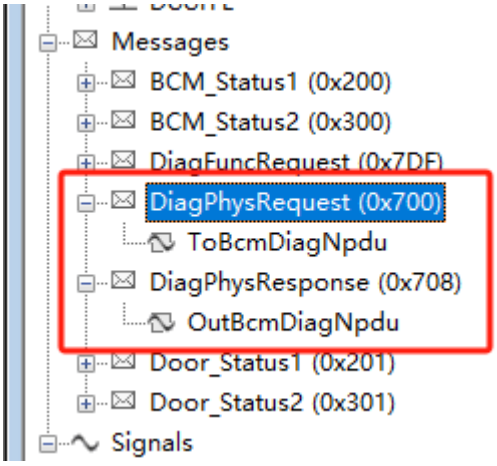
添加诊断Rx报文信号的定义



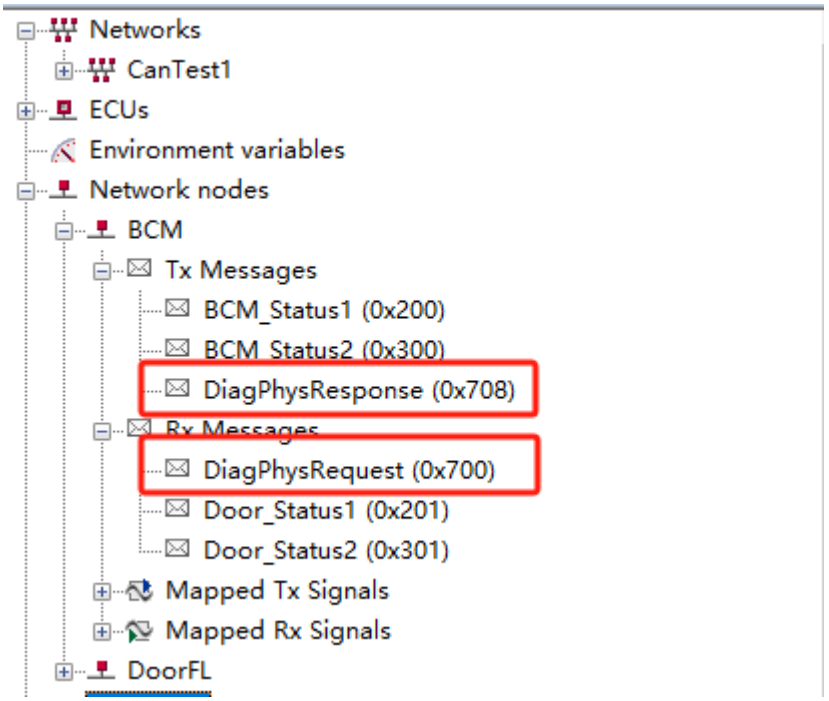
添加诊断Tx报文信号的定义



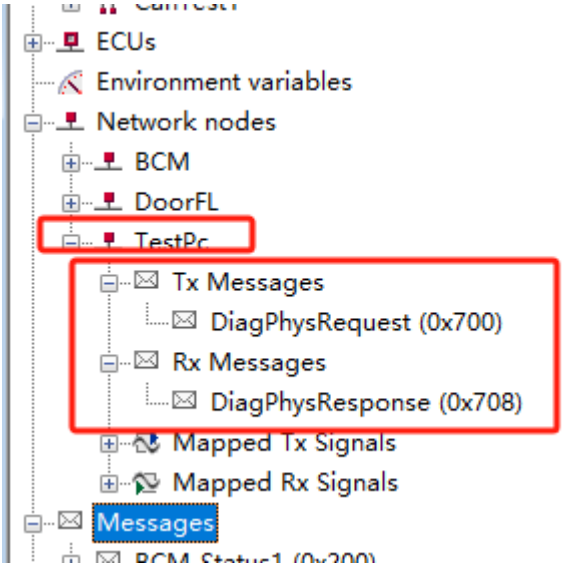
添加物理寻址 请求报文 0x700 和应答报文 0x708, 并把上述的信号映射到报文上



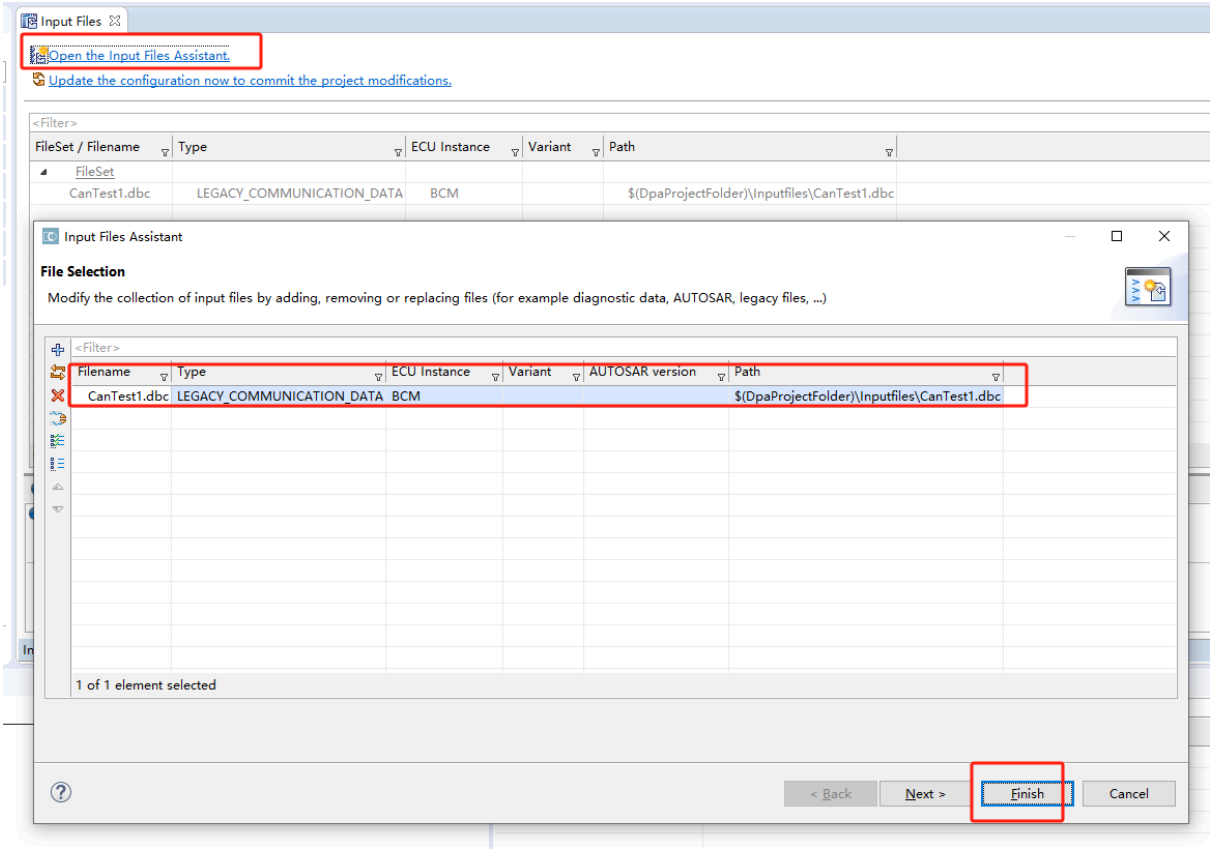
Bcm 节点（我们），添加Tx和Rx的报文



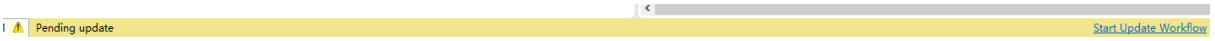
新建TestPc 节点，代表我们上位机，同样把报文添加到TestPc对应的Tx和Rx



二、导入（更新）带诊断报文DBC到davinci

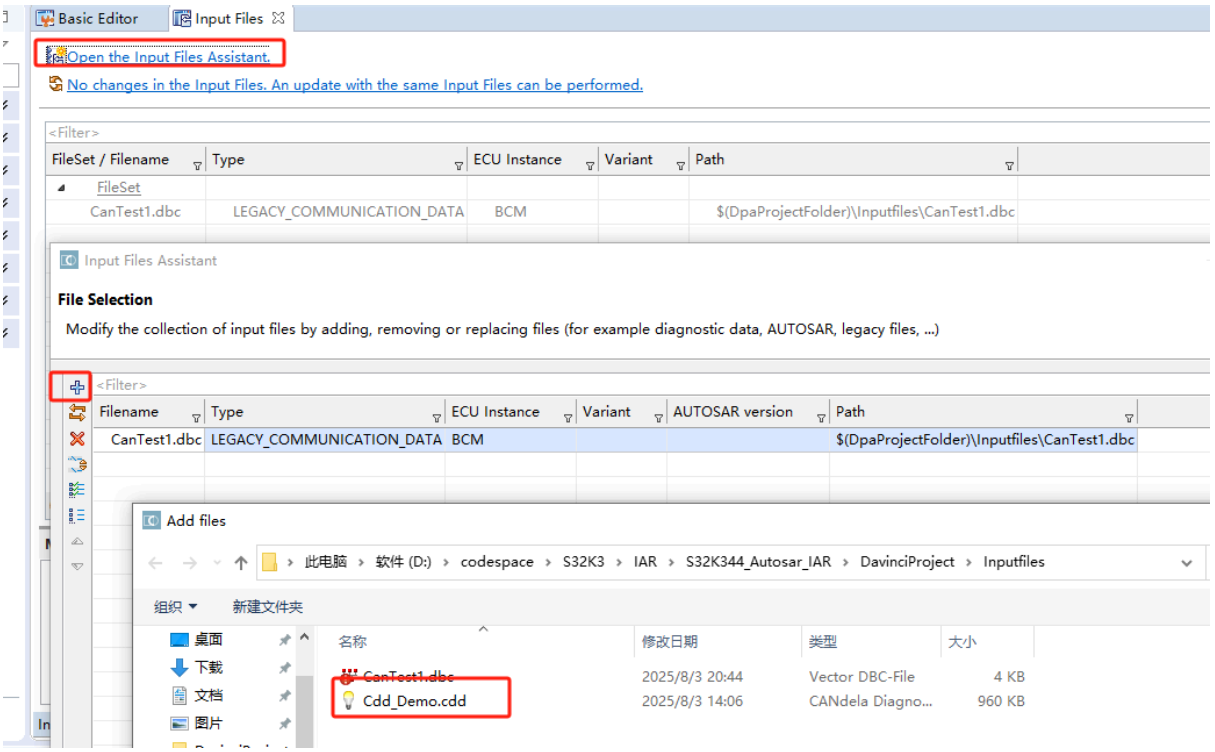


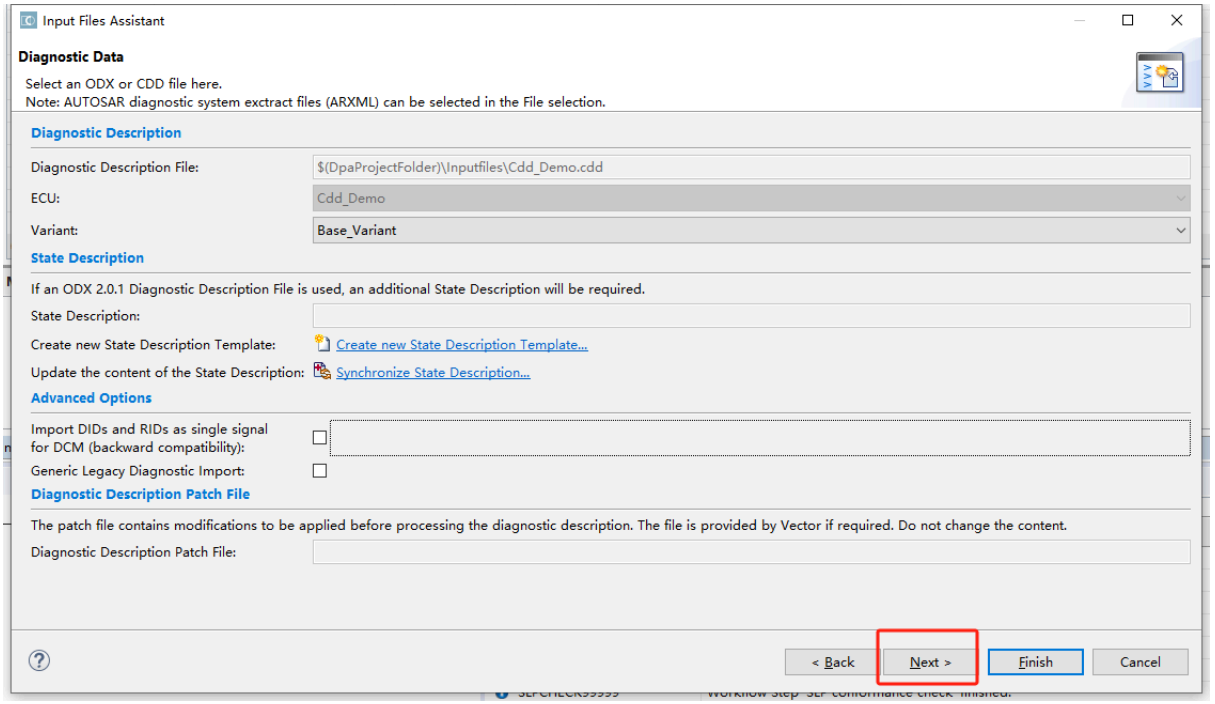
按底下的update

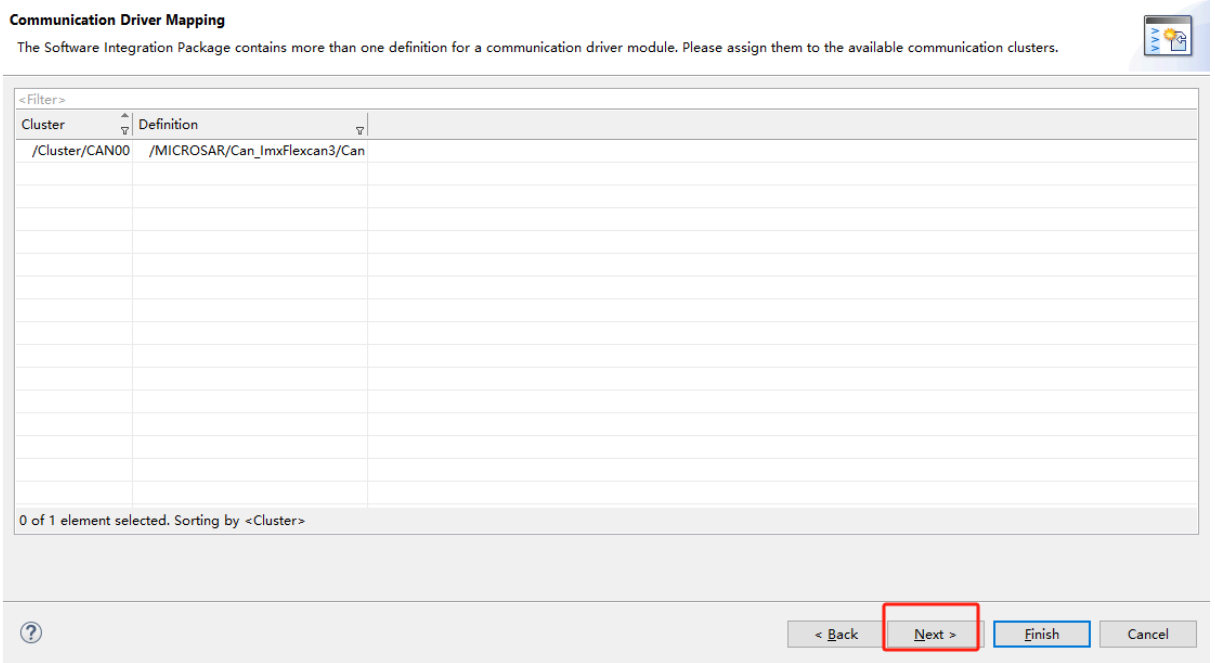


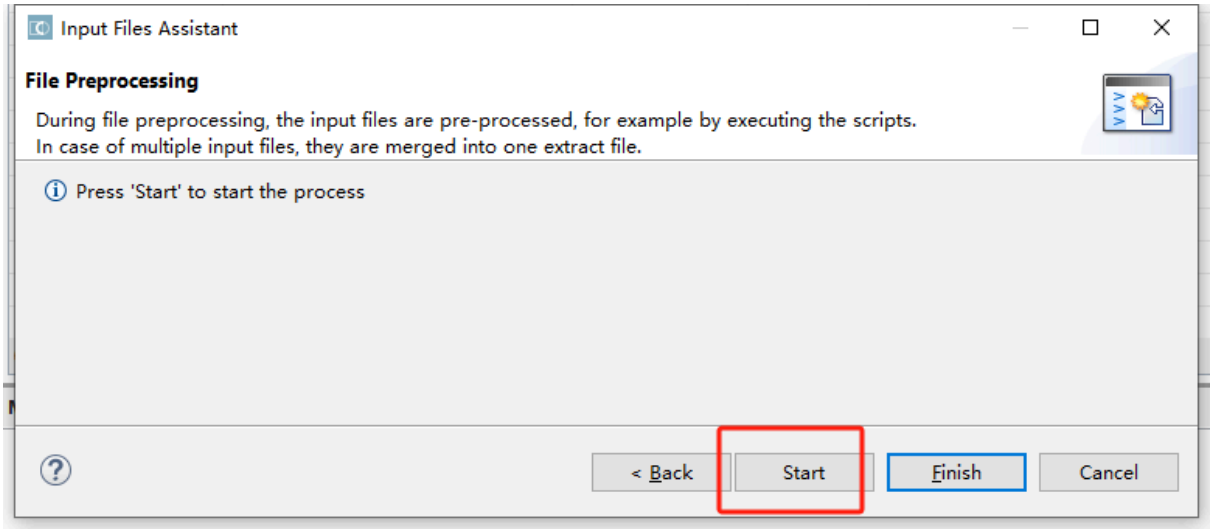
Input Files Update
Update Configuration
Press 'Update' to start the process.
Update Options
Complete update process
Update of ECU configuration and DaVinci Developer workspace will be performed.
Update ECU configuration only
DaVinci Developer workspace update will be skipped. Select this option only if the changes in the input files don't affect the DaVinci Developer workspace.
Update Legacy Diagnostic Data files only
For the Ecu Configuration update only the Legacy Diagnostic Data will be considered.
Update
Close

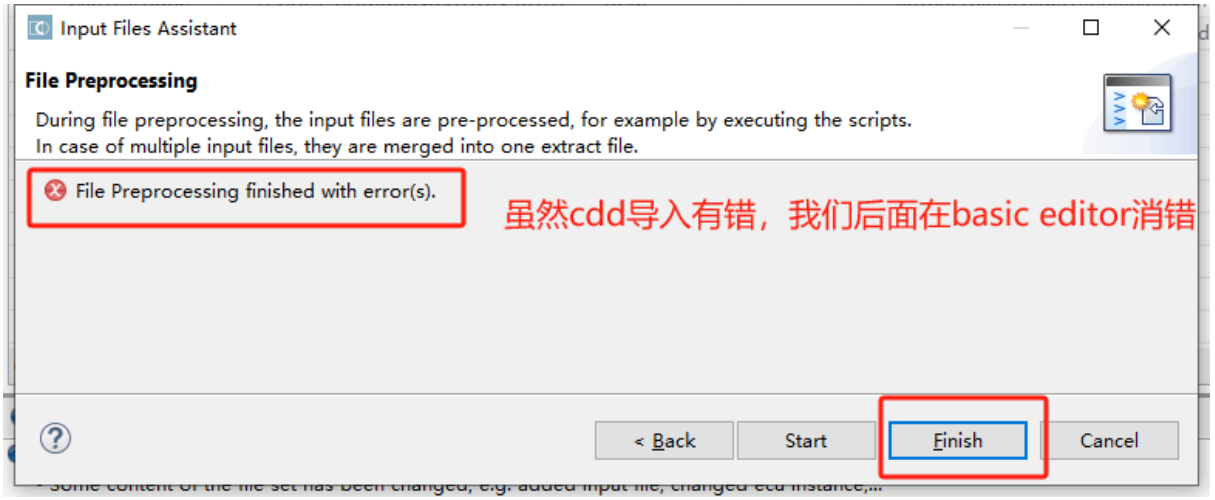
三、导入cdd文件


Basic Editor
Input Files
Open the Input Files Assistant.
No changes in the Input Files. An update with the same Input Files can be performed.
FileSet / Filename
Type
ECU Instance
Variant
Path
FileSet
CanTest1.dbc
LEGACY_COMMUNICATION_DATA
BCM
$(DpaProjectFolder)\Inputfiles\CanTest1.dbc
Input Files Assistant
File Selection
Modify the collection of input files by adding, removing or replacing files (for example diagnostic data, AUTOSAR, legacy files, ...)
Filename
AUTOSAR version
Add files
此电脑 › 软件 (D:) › codespace › S32K3 › IAR › S32K344_Autosar_IAR › DavinciProject › Inputfiles
组织
新建文件夹
桌面
下载
文档
图片
名称
修改日期
类型
大小
CanTest1.dbc
2025/8/3 20:44
Vector DBC-File
4 KB
Cdd_Demo.cdd
2025/8/3 14:06
CANdela Diagno...
960 KB

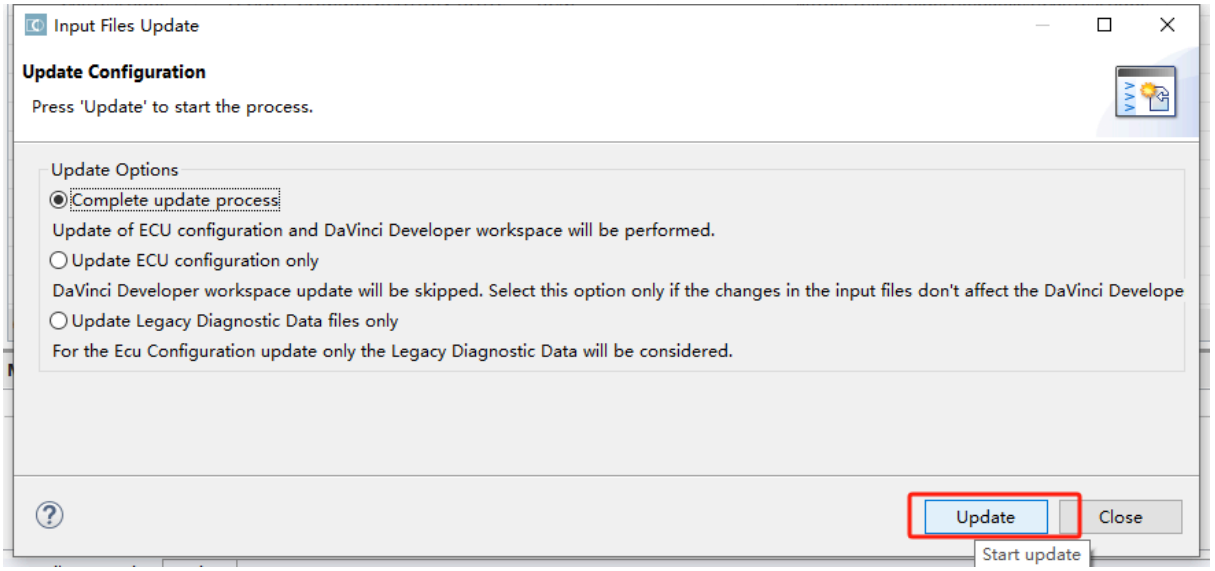

Input Files Assistant
Diagnostic Data
Select an ODX or CDD file here.
Note: AUTOSAR diagnostic system exctract files (ARXML) can be selected in the File selection.
Diagnostic Description
Diagnostic Description File:
$(DpaProjectFolder)\Inputfiles\Cdd_Demo.cdd
ECU:
Cdd_Demo
Variant:
Base_Variant
State Description
If an ODX 2.0.1 Diagnostic Description File is used, an additional State Description will be required.
State Description:
Create new State Description Template:
Create new State Description Template...
Update the content of the State Description:
Synchronize State Description...
Advanced Options
Import DIDs and RIDs as single signal for DCM (backward compatibility):
Generic Legacy Diagnostic Import:
Diagnostic Description Patch File
The patch file contains modifications to be applied before processing the diagnostic description. The file is provided by Vector if required. Do not change the content.
Diagnostic Description Patch File:
< Back
Next >
Finish
Cancel


Communication Driver Mapping
The Software Integration Package contains more than one definition for a communication driver module. Please assign them to the available communication clusters.
<Filter>
Cluster
Definition
/Cluster/CAN00
/MICROSAR/Can_ImxFlexcan3/Can
0 of 1 element selected. Sorting by <Cluster>
< Back
Next >
Finish
Cancel



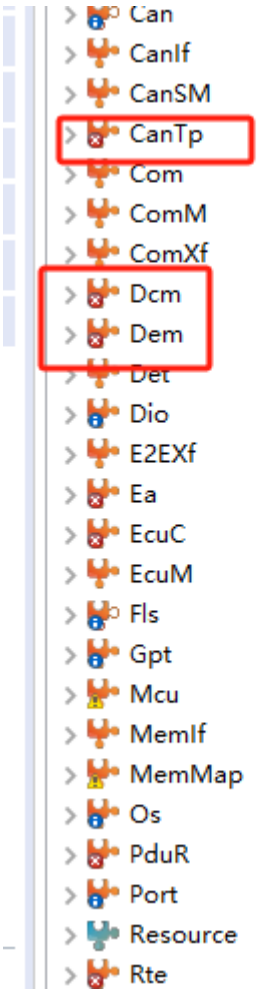


按底下黄色更新update

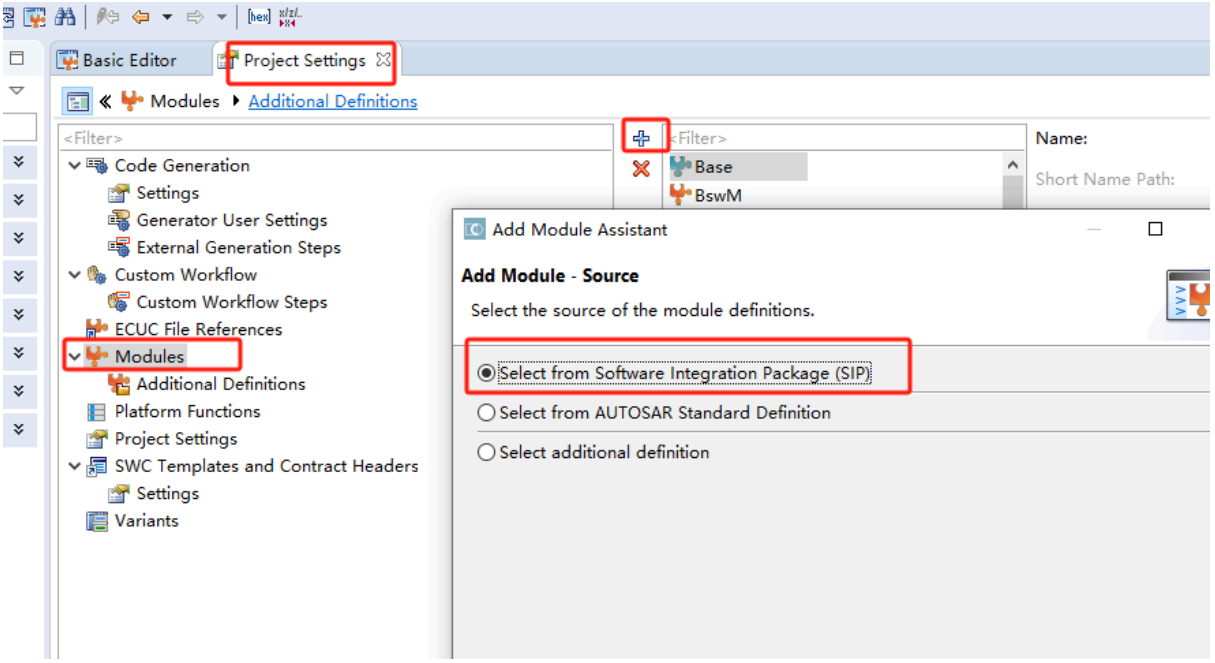


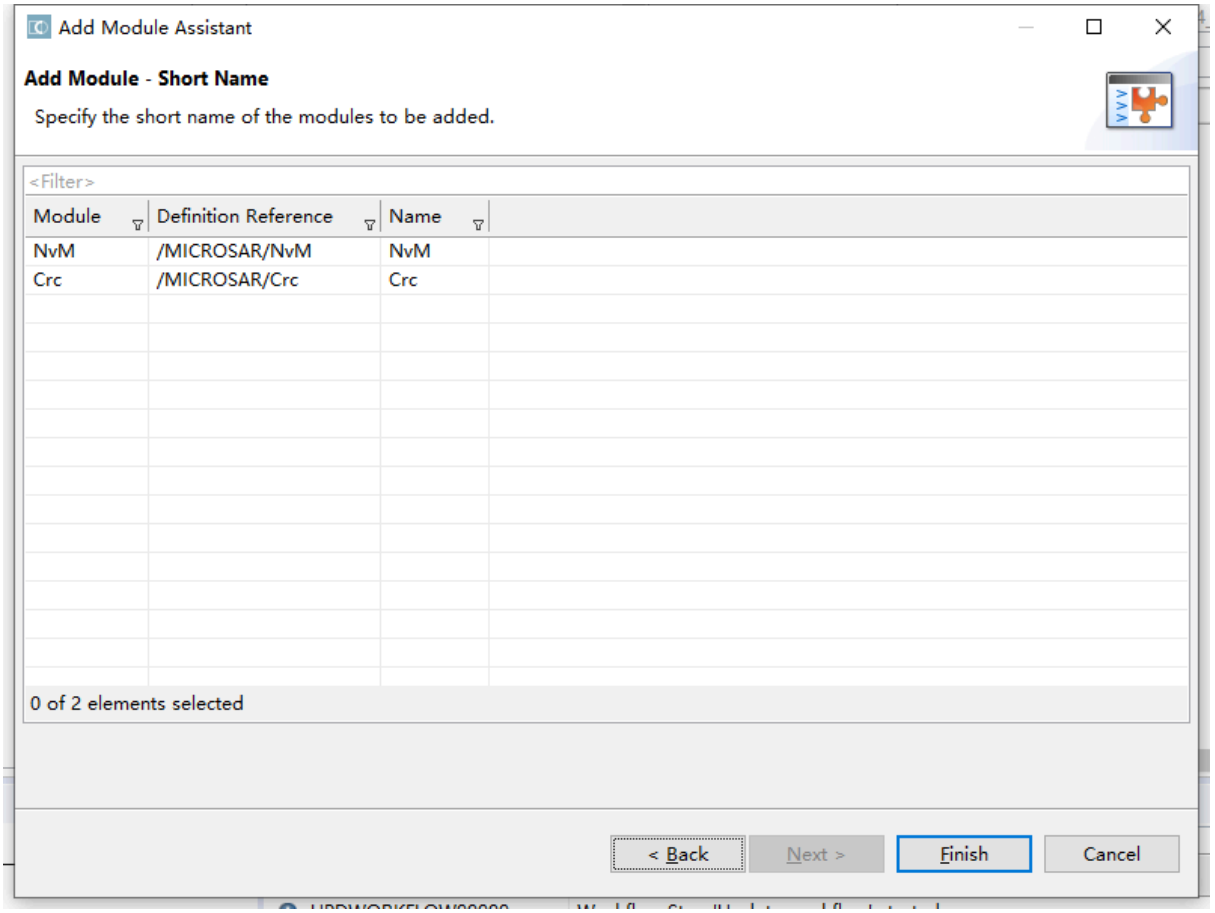
四、导入Nvm\Crc，并对所有模块消错

导入dbc和cdd后自动生成了CanTp、Dcm、Dem

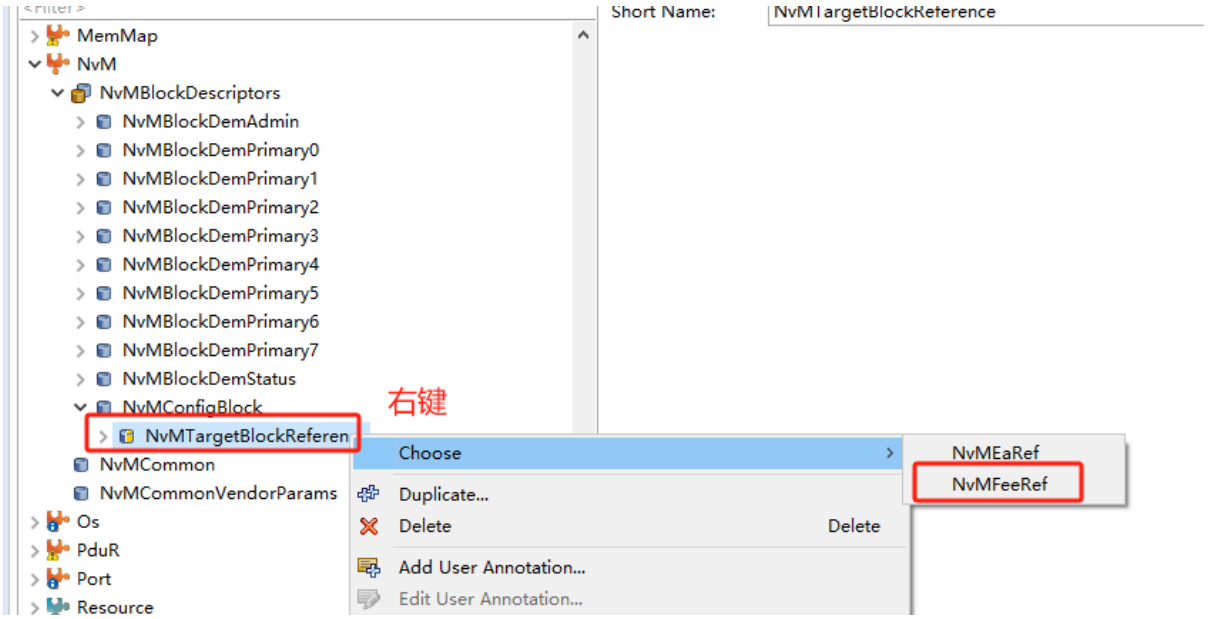


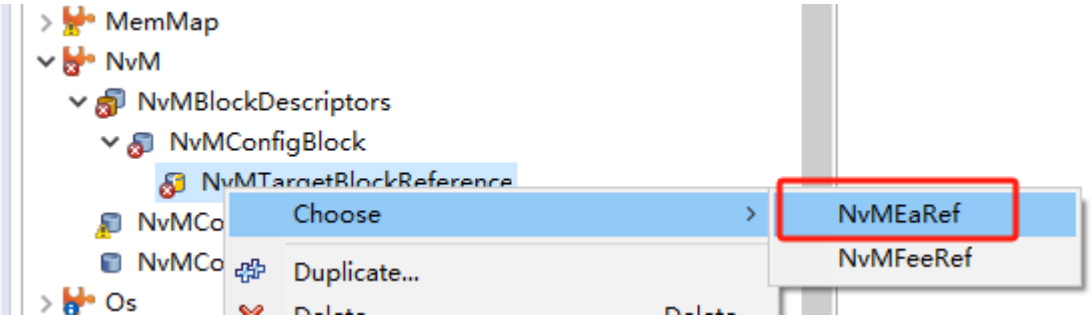
我们补充导入Nvm、Crc即可



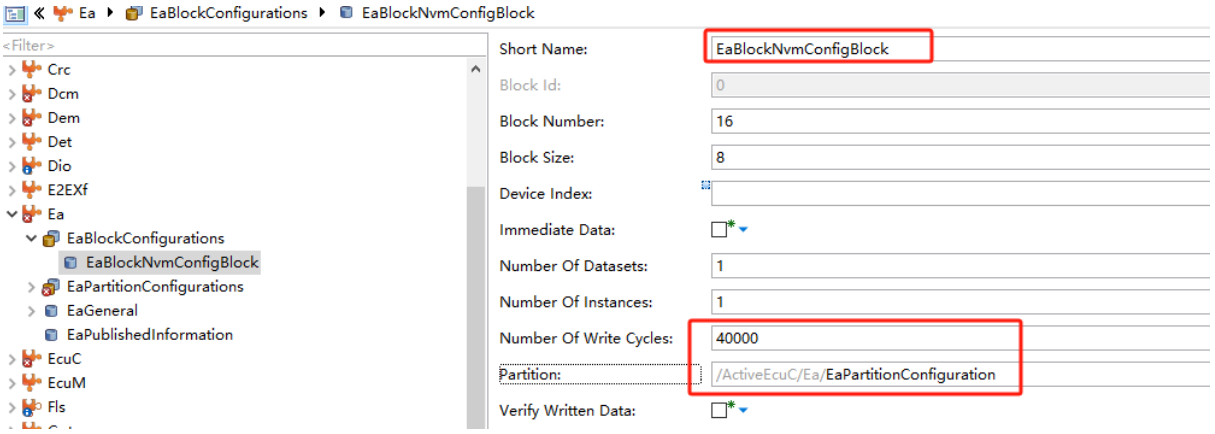


4.1、NvM模块消错和新增跟Dem相关的块

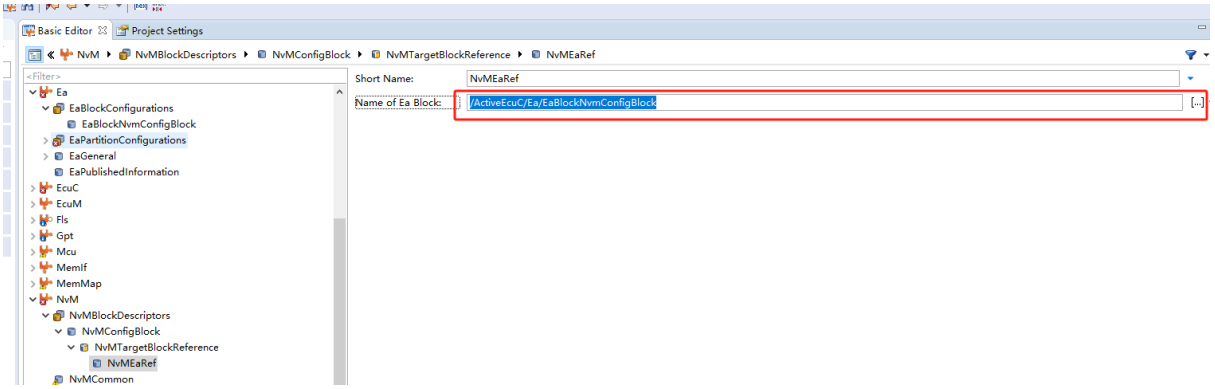




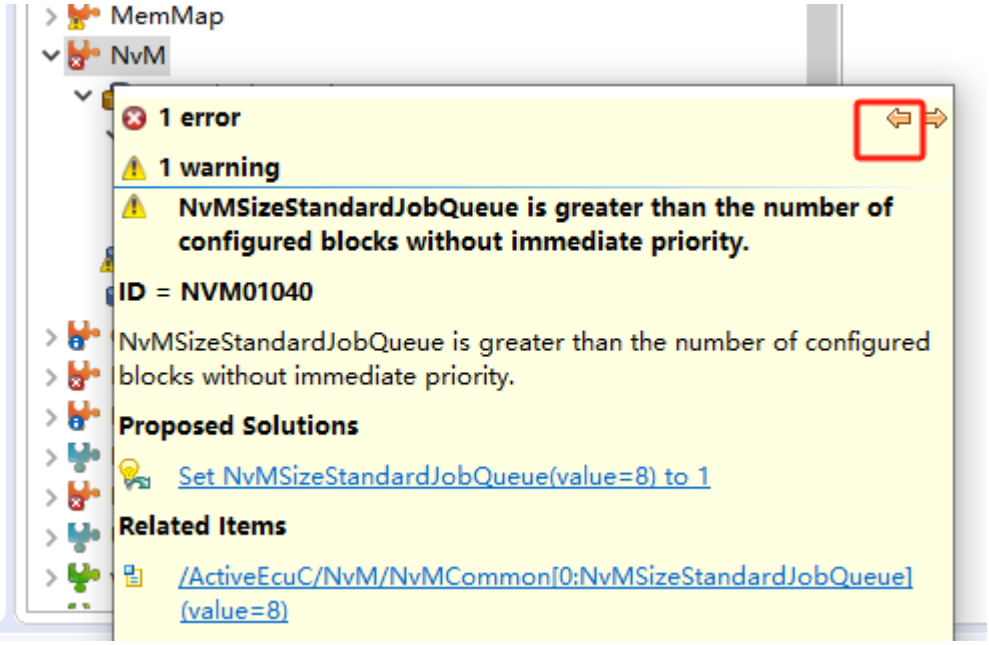
Ea我们只新建block给Nvm用，后面是不生成代码的



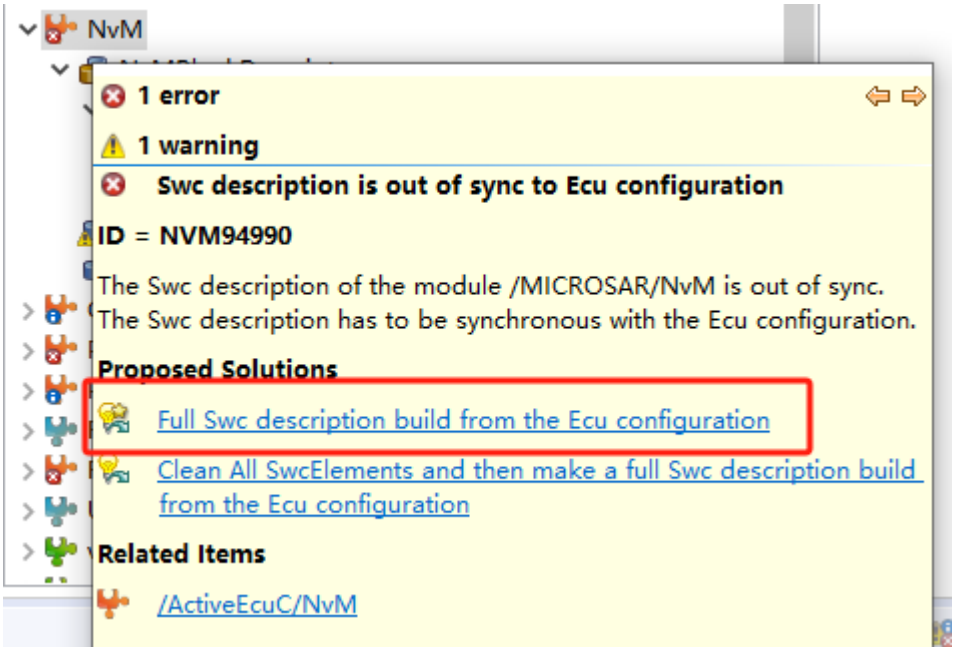
NvMConfigBlock选择的NvMEaRef选择上述的Ea block



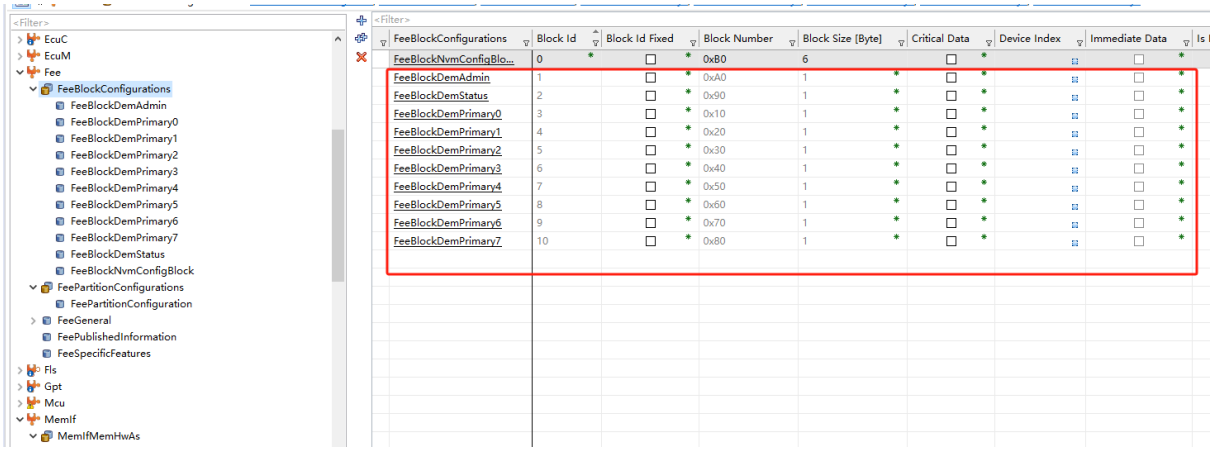
找到Nvm的swc同步



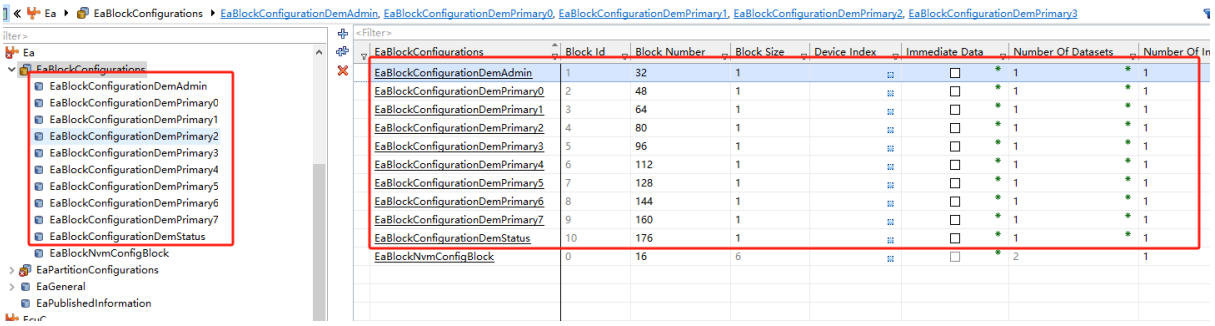
要点下面的两次，NvM模块才会消错完成



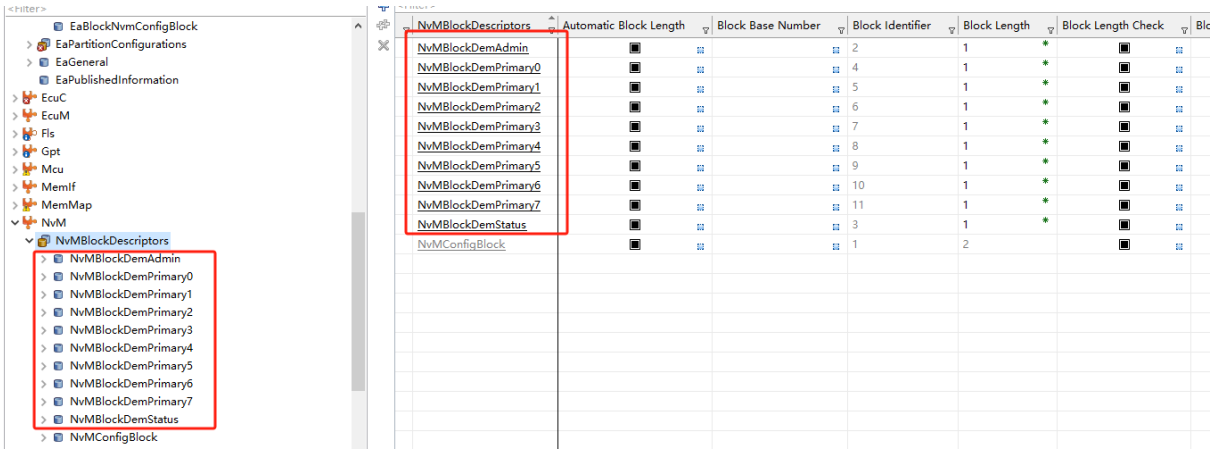
Fee添加Dem需要用到的块

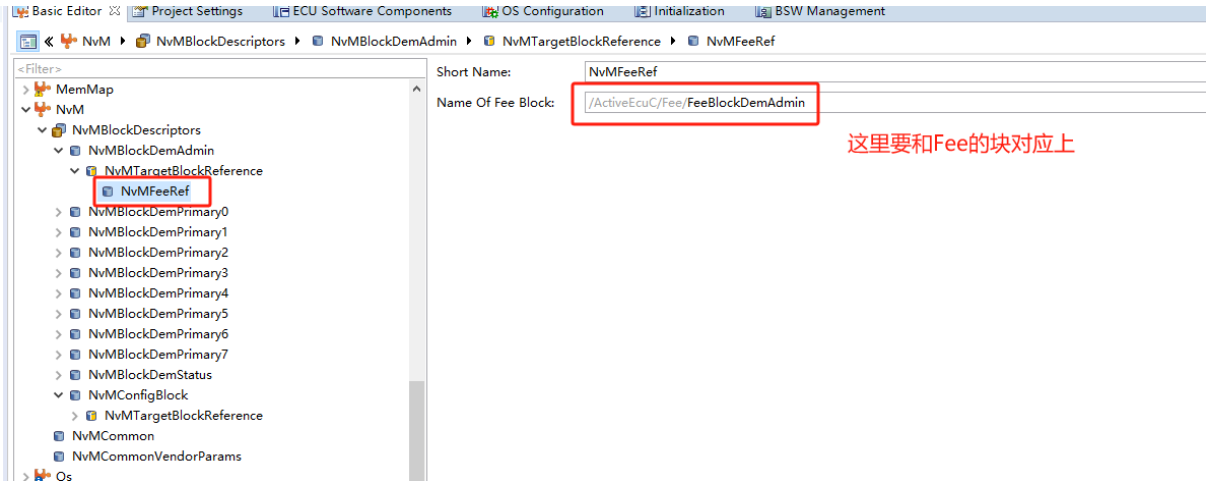


Ea 添加Dem需要用到的块



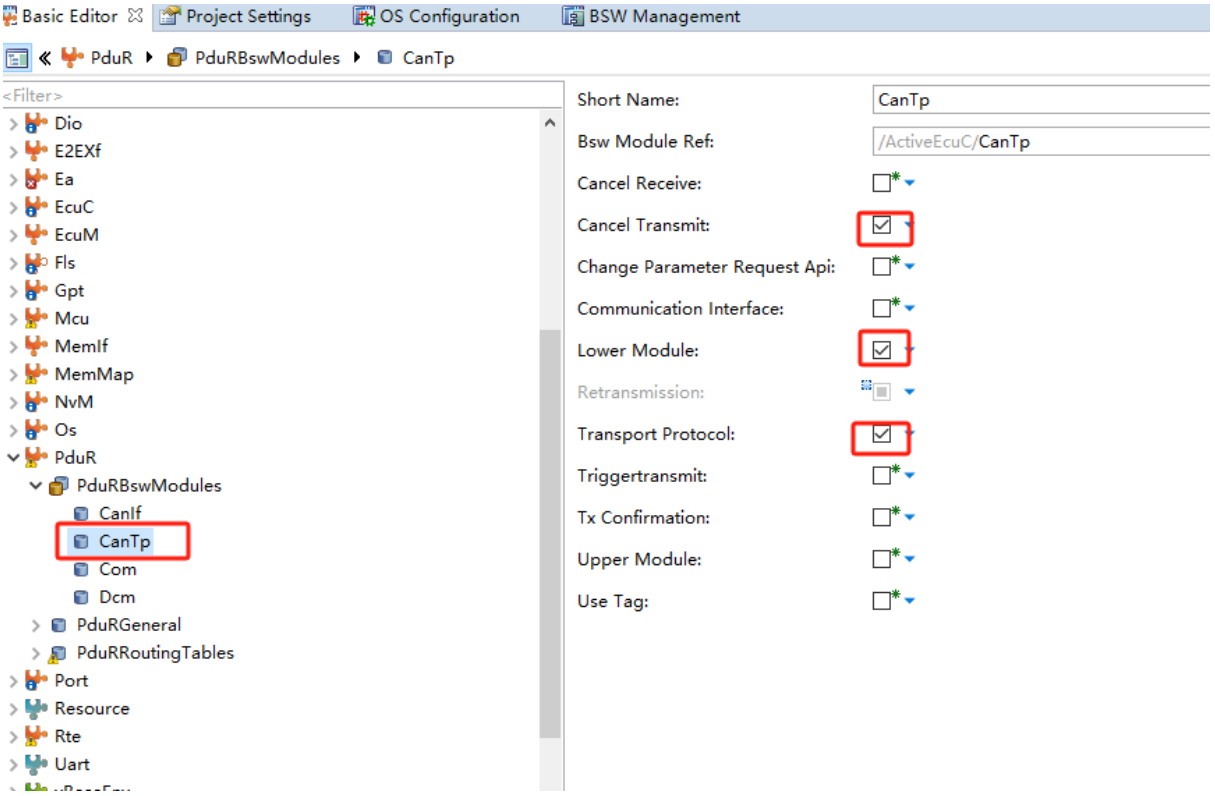
同样NvM也添加Dem需要用的块，并且和Fee的选择对应上

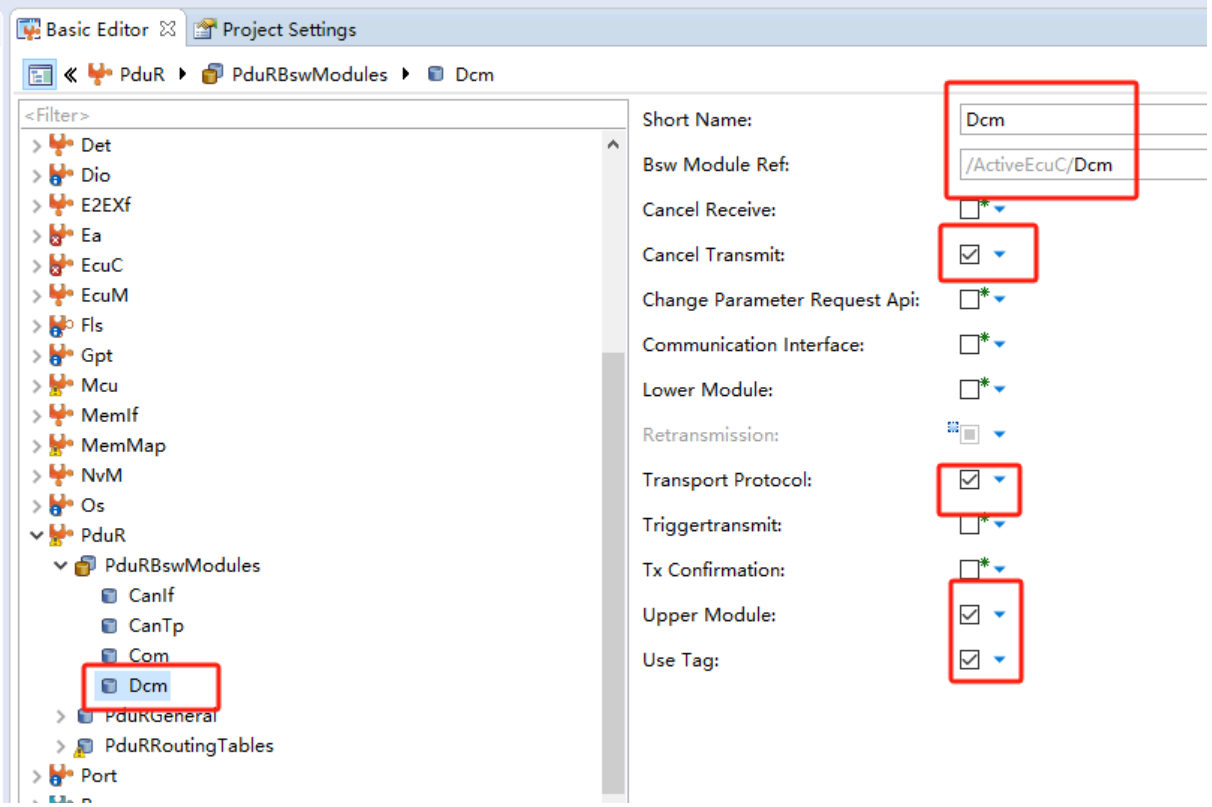




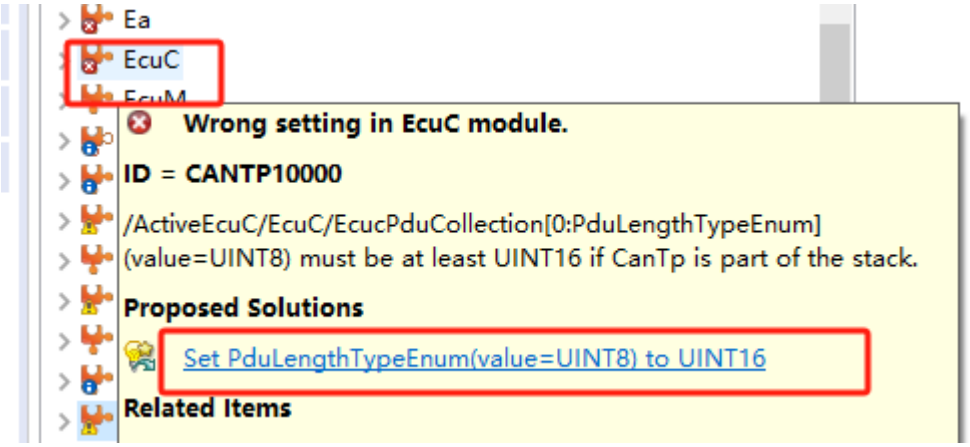
4.2、PduR模块消错

PduRBswModules新建 CanTp和Dcm

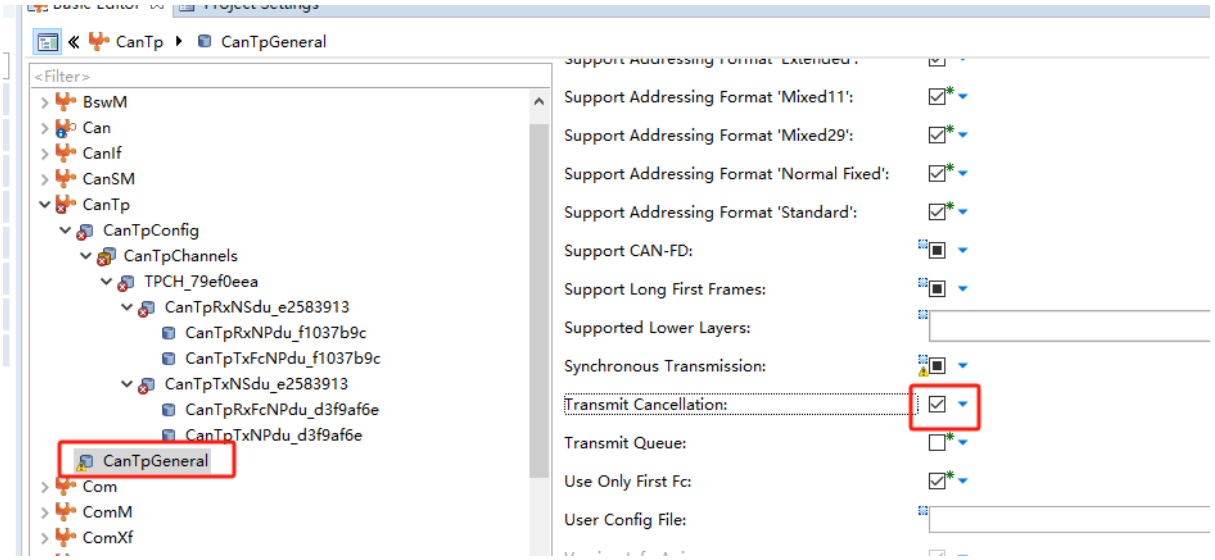

Basic Editor
Project Settings
OS Configuration
BSW Management
PduR
PduRBswModules
CanTp
<Filter>
Dio
E2EXf
Ea
EcuC
EcuM
Fls
Gpt
Mcu
MemIf
MemMap
NvM
Os
PduR
PduRBswModules
CanIf
CanTp
Com
Dcm
PduRGeneral
PduRRoutingTables
Port
Resource
Rte
Uart
Short Name:
CanTp
Bsw Module Ref:
/ActiveEcuC/CanTp
Cancel Receive:
Cancel Transmit:
Change Parameter Request Api:
Communication Interface:
Lower Module:
Retransmission:
Transport Protocol:
Triggertransmit:
Tx Confirmation:
Upper Module:
Use Tag:

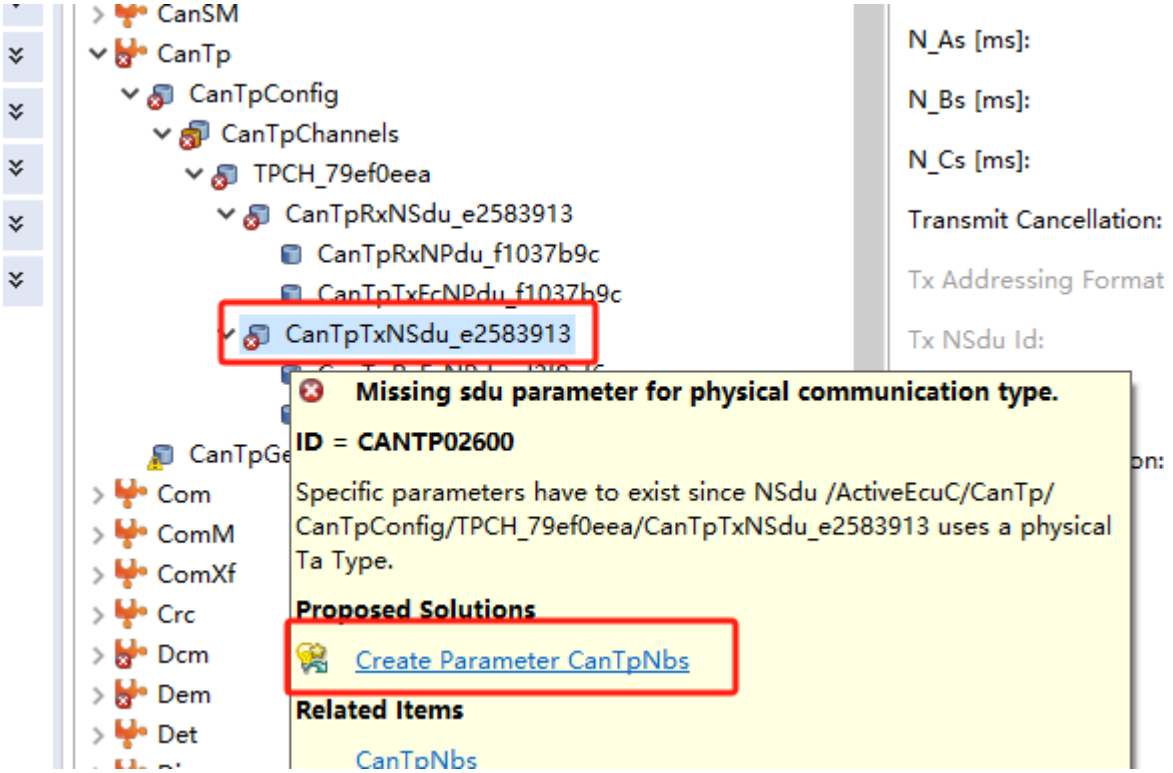

Basic Editor
Project Settings
PduR
PduRBswModules
Dcm
<Filter>
Det
Dio
E2EXf
Ea
EcuC
EcuM
Fls
Gpt
Mcu
MemIf
MemMap
NvM
Os
PduR
PduRBswModules
CanIf
CanTp
Com
Dcm
PduRGeneral
PduRRoutingTables
Port
Short Name:
Dcm
Bsw Module Ref:
/ActiveEcuC/Dcm
Cancel Receive:
Cancel Transmit:
Change Parameter Request Api:
Communication Interface:
Lower Module:
Retransmission:
Transport Protocol:
Triggertransmit:
Tx Confirmation:
Upper Module:
Use Tag:

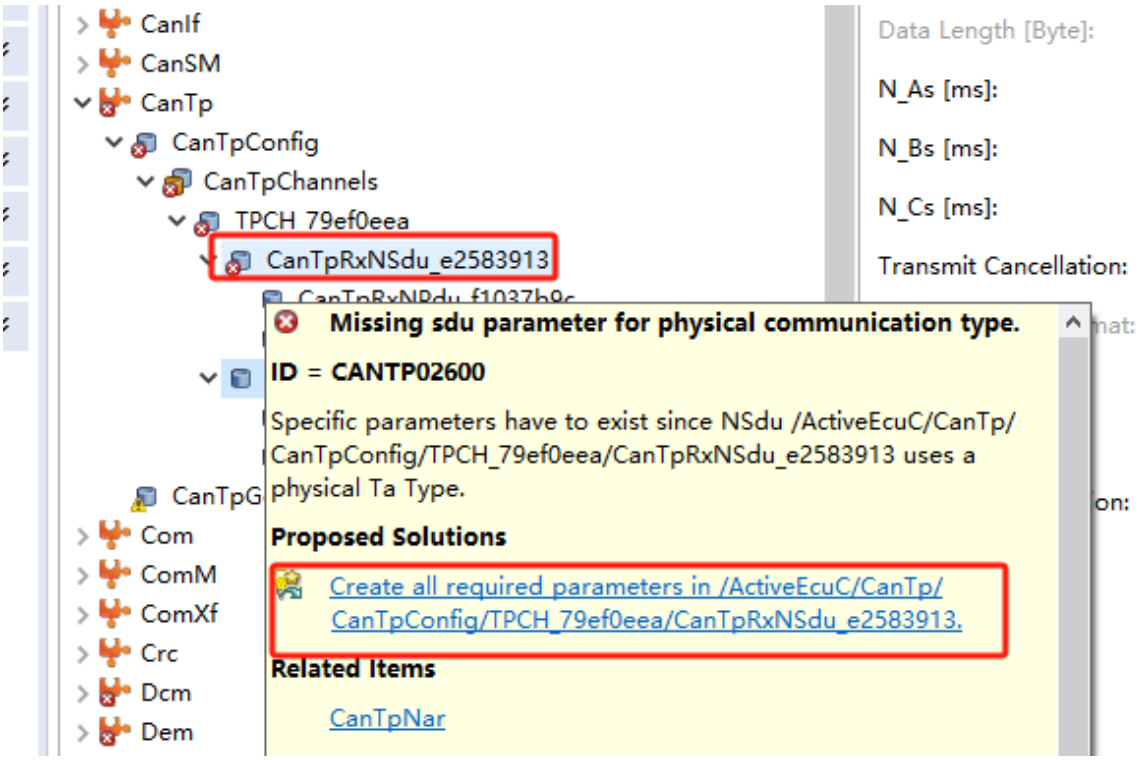
4.3、EcuC模块消错



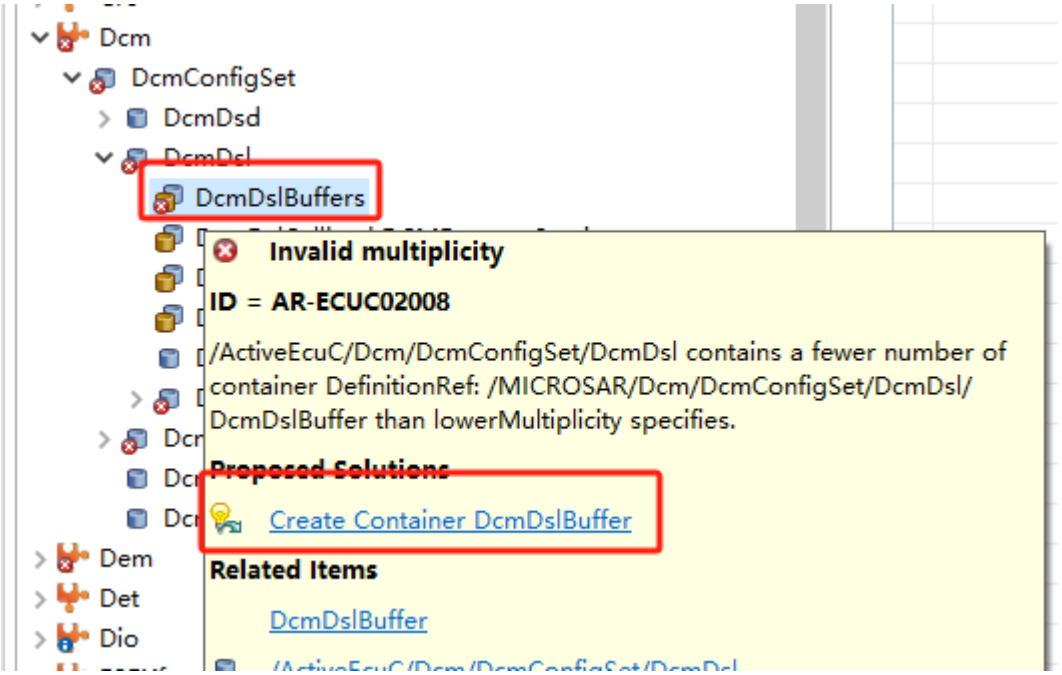
4.4、CanTp消错

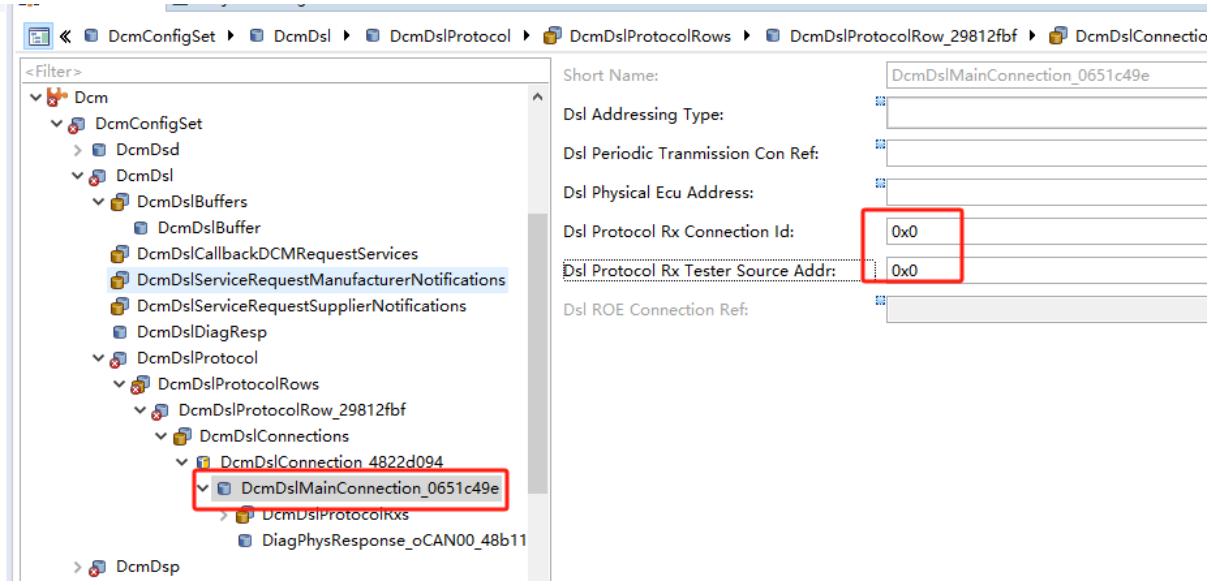


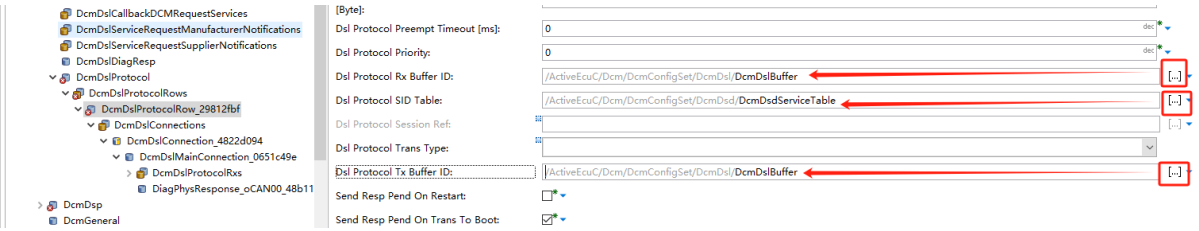

CanSM
CanTp
CanTpConfig
CanTpChannels
TPCH_79ef0eea
CanTpRxNSdu_e2583913
CanTpRxNPdu_f1037b9c
CanTpTxFcNPdu_f1037b9c
CanTpTxNSdu_e2583913
Com
ComM
ComXf
Crc
Dcm
Dem
Det
N_As [ms]:
N_Bs [ms]:
N_Cs [ms]:
Transmit Cancellation:
Tx Addressing Format
Tx NSdu Id:
Missing sdu parameter for physical communication type.
ID = CANTP02600
Specific parameters have to exist since NSdu /ActiveEcuC/CanTp/CanTpConfig/TPCH_79ef0eea/CanTpTxNSdu_e2583913 uses a physical Ta Type.
Proposed Solutions
Create Parameter CanTpNbs
Related Items
CanTpNbs

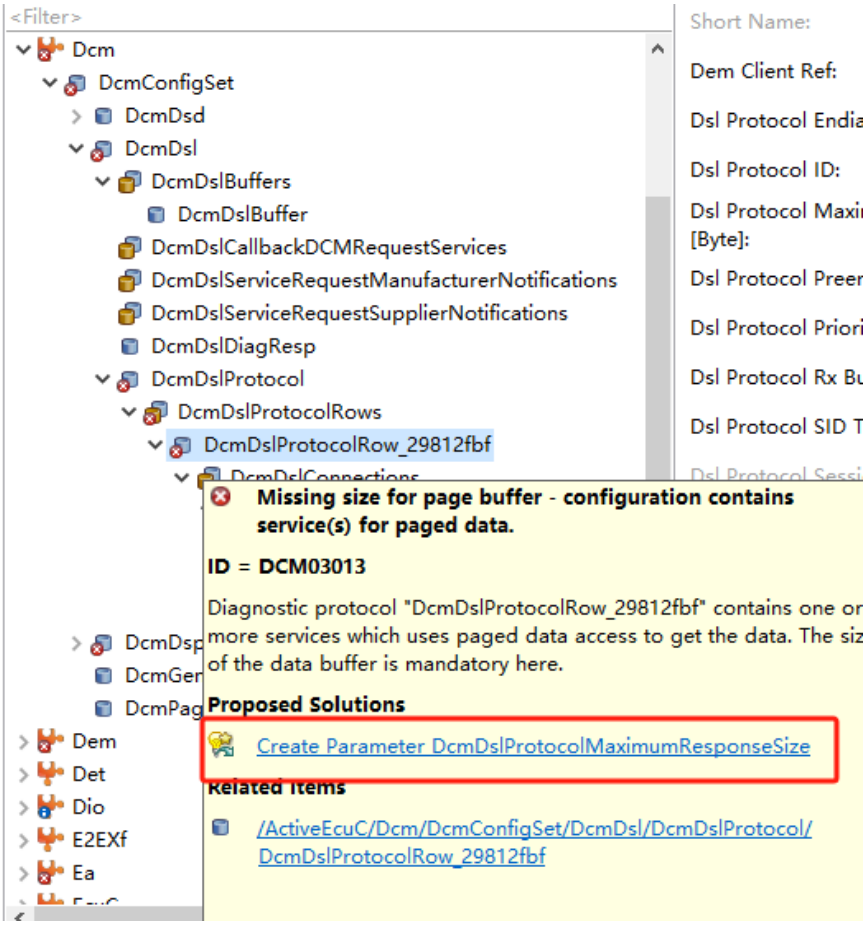

CanIf
CanSM
CanTp
CanTpConfig
CanTpChannels
TPCH_79ef0eea
CanTpRxNSdu_e2583913
Com
ComM
ComXf
Crc
Dcm
Dem
Data Length [Byte]:
N_As [ms]:
N_Bs [ms]:
N_Cs [ms]:
Transmit Cancellation:
Missing sdu parameter for physical communication type.
ID = CANTP02600
Specific parameters have to exist since NSdu /ActiveEcuC/CanTp/CanTpConfig/TPCH_79ef0eea/CanTpRxNSdu_e2583913 uses a physical Ta Type.
Proposed Solutions
Create all required parameters in /ActiveEcuC/CanTp/CanTpConfig/TPCH_79ef0eea/CanTpRxNSdu_e2583913.
Related Items
CanTpNar

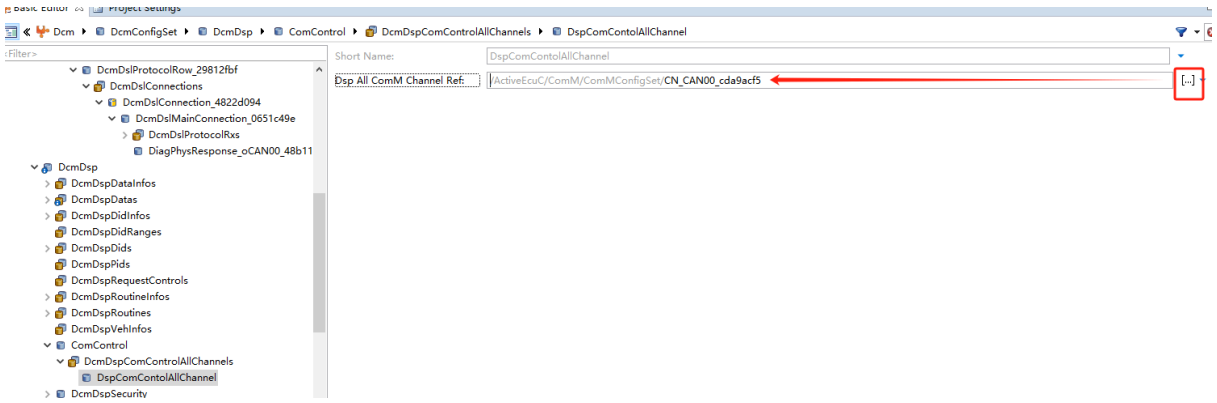
4.5、Dcm模块消错


Dcm
DcmConfigSet
DcmDsd
DcmDsl
DcmDslBuffers
Invalid multiplicity
ID = AR-ECUC02008
/ActiveEcuC/Dcm/DcmConfigSet/DcmDsl contains a fewer number of container DefinitionRef: /MICROSAR/Dcm/DcmConfigSet/DcmDsl/DcmDslBuffer than lowerMultiplicity specifies.
Proposed Solutions
Create Container DcmDslBuffer
Related Items
DcmDslBuffer
Dem
Det
Dio

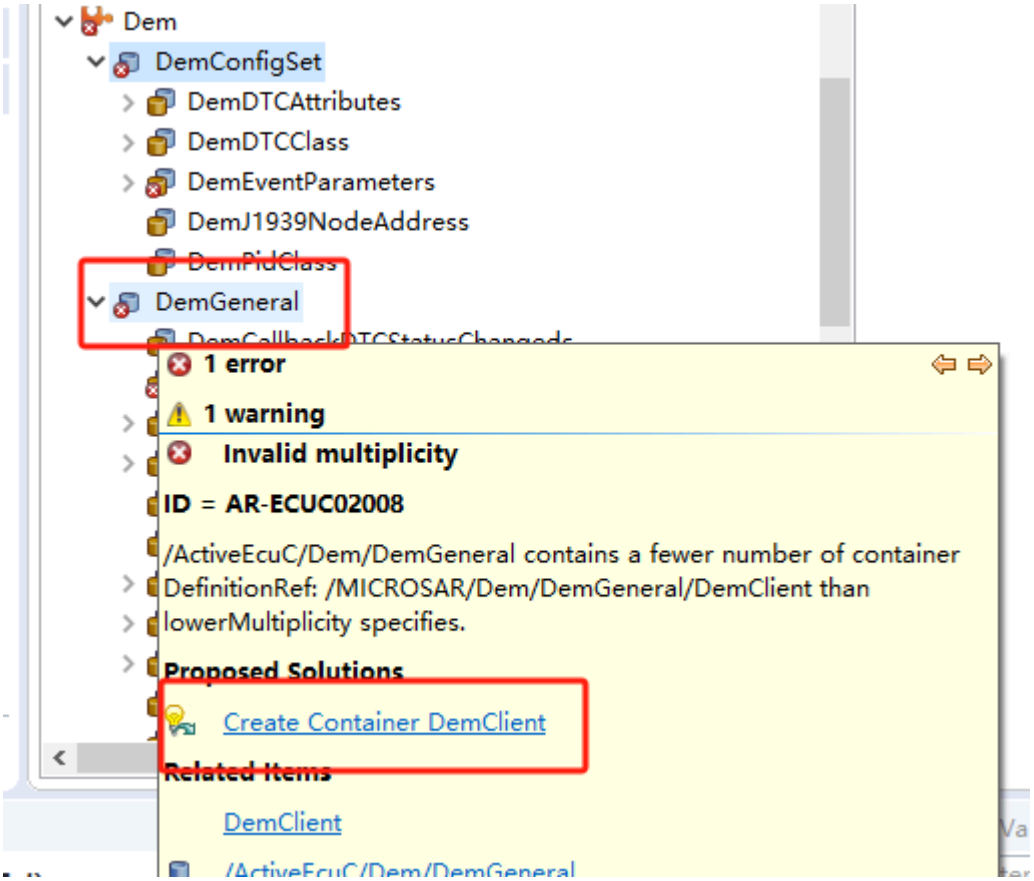

DcmConfigSet
DcmDsl
DcmDslProtocol
DcmDslProtocolRows
DcmDslProtocolRow_29812fbf
DcmDslConnectio
<Filter>
Dcm
DcmConfigSet
DcmDsd
DcmDsl
DcmDslBuffers
DcmDslBuffer
DcmDslCallbackDCMRequestServices
DcmDslServiceRequestManufacturerNotifications
DcmDslServiceRequestSupplierNotifications
DcmDslDiagResp
DcmDslProtocol
DcmDslProtocolRows
DcmDslProtocolRow_29812fbf
DcmDslConnections
DcmDslConnection_4822d094
DcmDslMainConnection_0651c49e
DcmDslProtocolRxs
DiagPhysResponse_oCAN00_48b11
DcmDsp
Short Name:
DcmDslMainConnection_0651c49e
Dsl Addressing Type:
Dsl Periodic Tranmission Con Ref:
Dsl Physical Ecu Address:
Dsl Protocol Rx Connection Id:
0x0
Dsl Protocol Rx Tester Source Addr:
0x0
Dsl ROE Connection Ref:

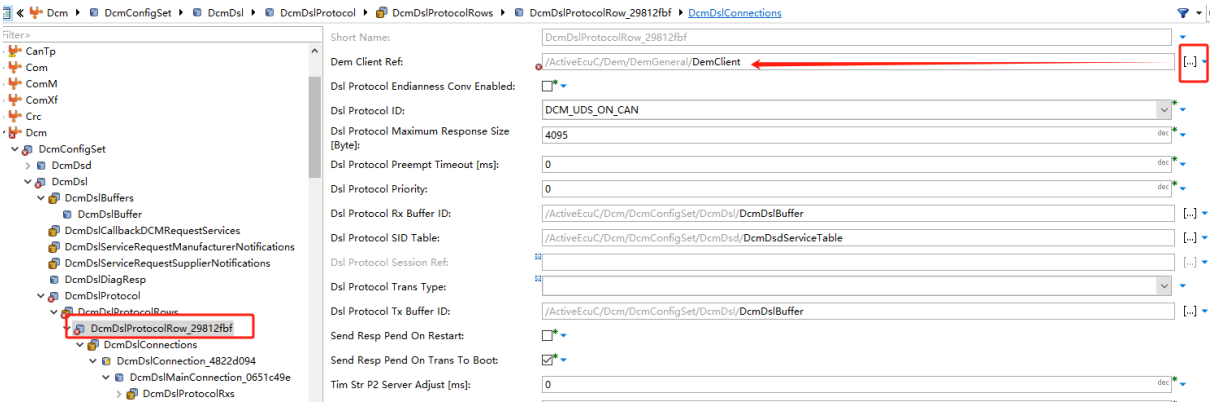

DcmDslCallbackDCMRequestServices
DcmDslServiceRequestManufacturerNotifications
DcmDslServiceRequestSupplierNotifications
DcmDslDiagResp
DcmDslProtocol
DcmDslProtocolRows
DcmDslProtocolRow_29812fbf
DcmDslConnections
DcmDslConnection_4822d094
DcmDslMainConnection_0651c49e
DcmDslProtocolRxs
DiagPhysResponse_oCAN00_48b11
DcmDsp
DcmGeneral
[Byte]:
Dsl Protocol Preempt Timeout [ms]:
0
Dsl Protocol Priority:
0
Dsl Protocol Rx Buffer ID:
/ActiveEcuC/Dcm/DcmConfigSet/DcmDsl/DcmDslBuffer
Dsl Protocol SID Table:
/ActiveEcuC/Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable
Dsl Protocol Session Ref:
Dsl Protocol Trans Type:
Dsl Protocol Tx Buffer ID:
/ActiveEcuC/Dcm/DcmConfigSet/DcmDsl/DcmDslBuffer
Send Resp Pend On Restart:
Send Resp Pend On Trans To Boot:

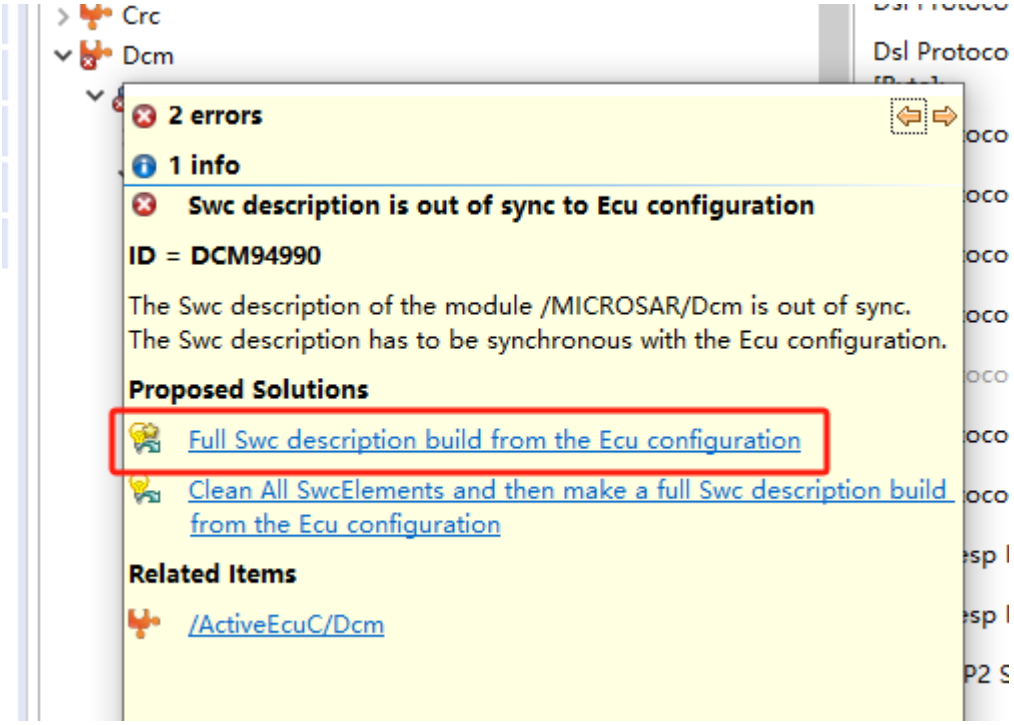

Dcm
DcmConfigSet
DcmDsd
DcmDsl
DcmDslBuffers
DcmDslBuffer
DcmDslCallbackDCMRequestServices
DcmDslServiceRequestManufacturerNotifications
DcmDslServiceRequestSupplierNotifications
DcmDslDiagResp
DcmDslProtocol
DcmDslProtocolRows
DcmDslProtocolRow_29812fbf
Dem
Det
Dio
E2EXf
Ea
Short Name:
Dem Client Ref:
Dsl Protocol ID:
Missing size for page buffer - configuration contains service(s) for paged data.
ID = DCM03013
Diagnostic protocol "DcmDslProtocolRow_29812fbf" contains one or more services which uses paged data access to get the data. The siz of the data buffer is mandatory here.
Proposed Solutions
Create Parameter DcmDslProtocolMaximumResponseSize
Related Items
/ActiveEcuC/Dcm/DcmConfigSet/DcmDsl/DcmDslProtocol/DcmDslProtocolRow_29812fbf


Dcm
DcmConfigSet
DcmDsp
ComControl
DcmDspComControlAllChannels
DspComContolAllChannel
Short Name:
Dsp All ComM Channel Ref:
/ActiveEcuC/ComM/ComMConfigSet/CN_CAN00_cda9acf5
DcmDslProtocolRow_29812fbf
DcmDslConnections
DcmDslConnection_4822d094
DcmDslMainConnection_0651c49e
DcmDslProtocolRxs
DiagPhysResponse_oCAN00_48b11
DcmDsp
DcmDspDataInfos
DcmDspDatas
DcmDspDidInfos
DcmDspDidRanges
DcmDspDids
DcmDspPids
DcmDspRequestControls
DcmDspRoutineInfos
DcmDspRoutines
DcmDspVehInfos
ComControl
DcmDspComControlAllChannels
DspComContolAllChannel
DcmDspSecurity

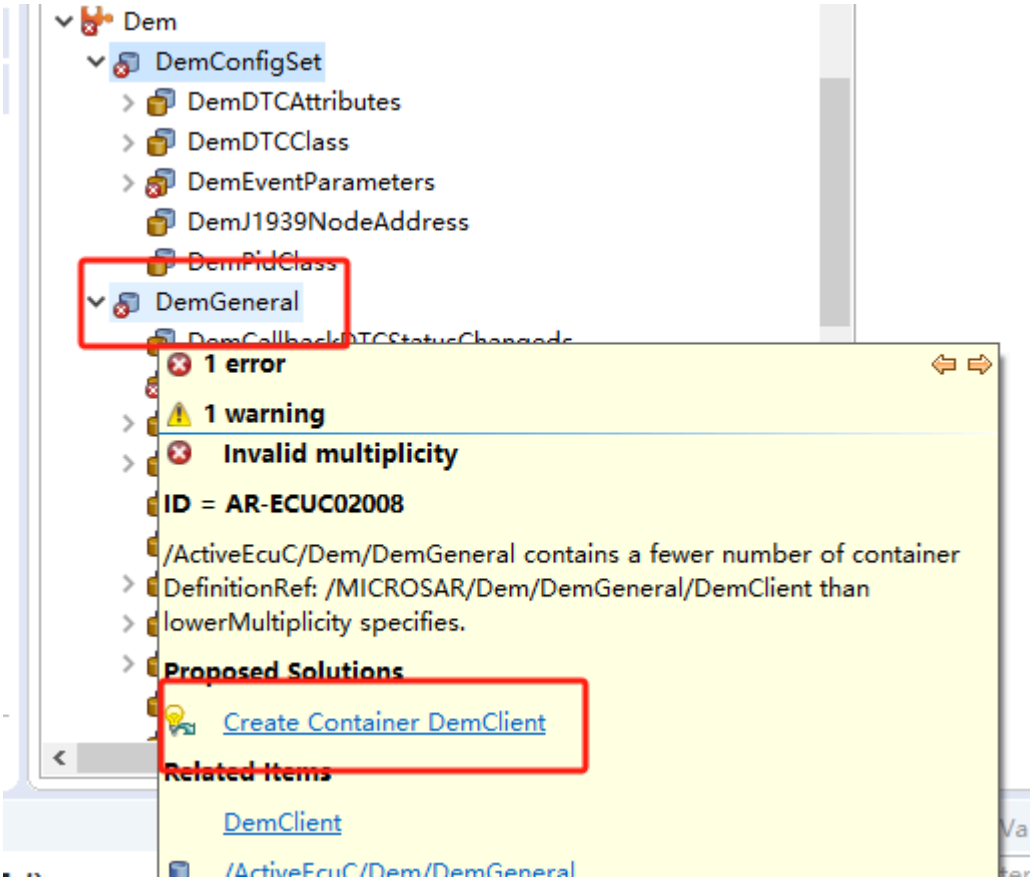
先Dem


Dem
DemConfigSet
DemDTCAttributes
DemDTCClass
DemEventParameters
DemJ1939NodeAddress
DemPidClass
DemGeneral
1 error
1 warning
Invalid multiplicity
ID = AR-ECUC02008
/ActiveEcuC/Dem/DemGeneral contains a fewer number of container DefinitionRef: /MICROSAR/Dem/DemGeneral/DemClient than lowerMultiplicity specifies.
Proposed Solutions
Create Container DemClient
Related Items
DemClient


Dcm
DcmConfigSet
DcmDsl
DcmDslProtocol
DcmDslProtocolRows
DcmDslProtocolRow_29812fbf
DcmDslConnections
Short Name:
Dem Client Ref:
/ActiveEcuC/Dem/DemGeneral/DemClient
Dsl Protocol Endianness Conv Enabled:
Dsl Protocol ID:
DCM_UDS_ON_CAN
Dsl Protocol Maximum Response Size [Byte]:
4095
Dsl Protocol Preempt Timeout [ms]:
Dsl Protocol Priority:
Dsl Protocol Rx Buffer ID:
/ActiveEcuC/Dcm/DcmConfigSet/DcmDsl/DcmDslBuffer
Dsl Protocol SID Table:
/ActiveEcuC/Dcm/DcmConfigSet/DcmDsd/DcmDsdServiceTable
Dsl Protocol Session Ref:
Dsl Protocol Trans Type:
Dsl Protocol Tx Buffer ID:
Send Resp Pend On Restart:
Send Resp Pend On Trans To Boot:
Tim Str P2 Server Adjust [ms]:

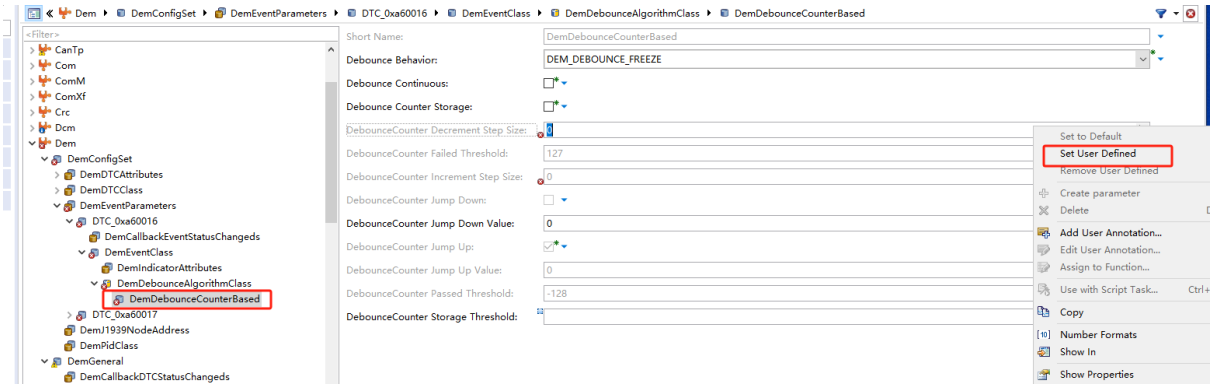

Crc
Dcm
Dsl Protoco
2 errors
1 info
Swc description is out of sync to Ecu configuration
ID = DCM94990
The Swc description of the module /MICROSAR/Dcm is out of sync.
The Swc description has to be synchronous with the Ecu configuration.
Proposed Solutions
Full Swc description build from the Ecu configuration
Clean All SwcElements and then make a full Swc description build from the Ecu configuration
Related Items
/ActiveEcuC/Dcm

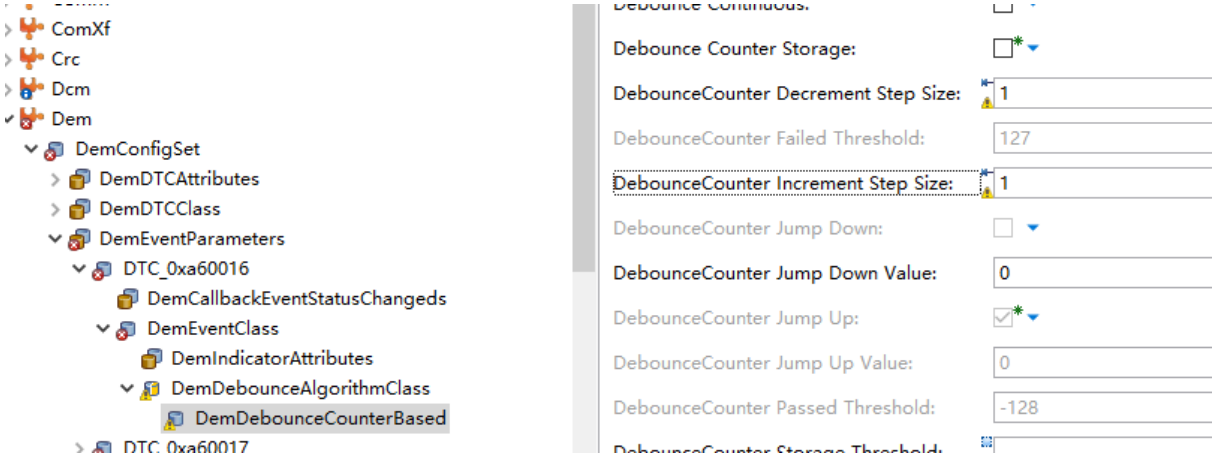
4.6、Dem消错

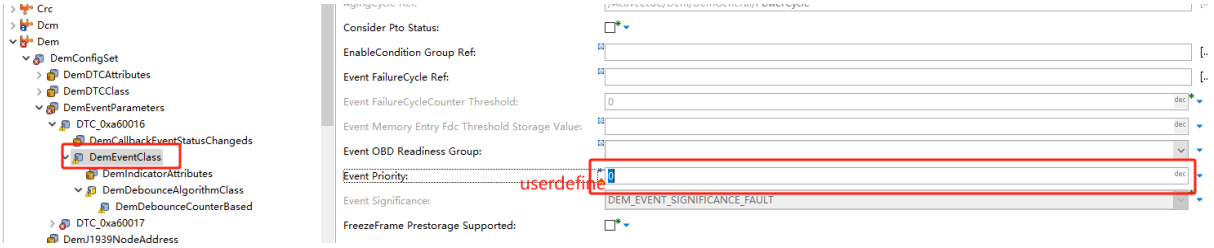


两个DTC一样的消错方式

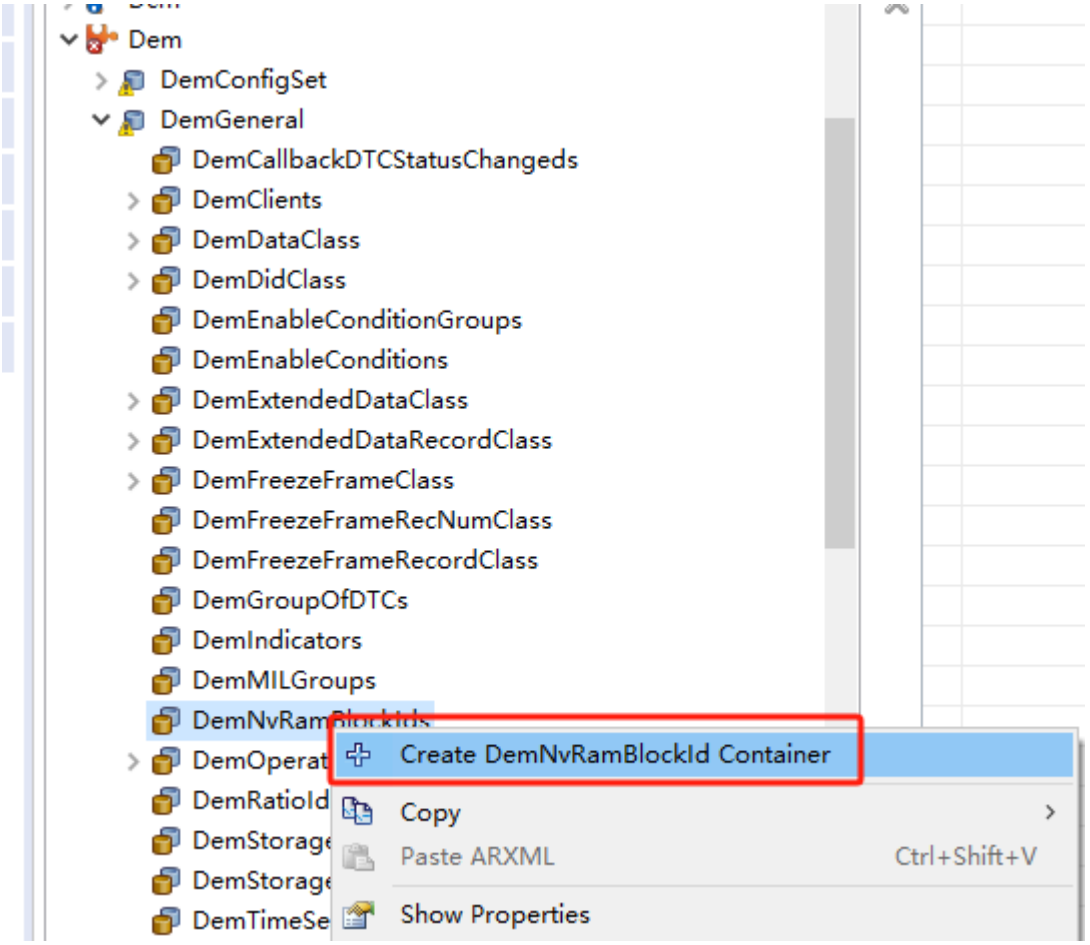
两个都改为User Defined, 数字改为1

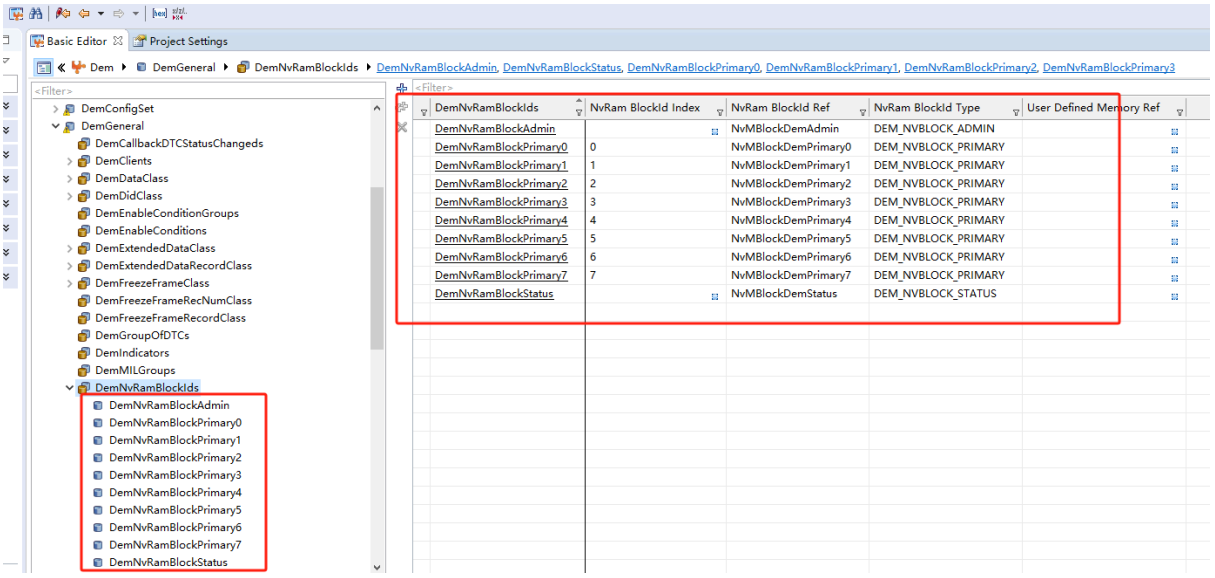




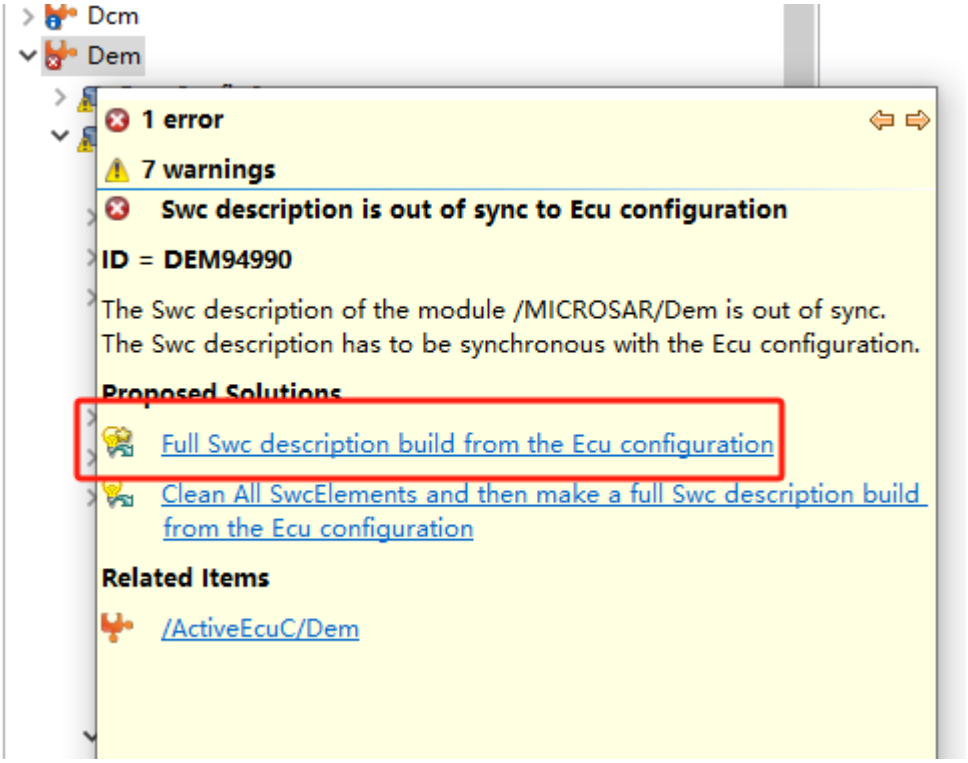


DTC_0xa60017跟上面的DTC一样的消错方式


Dem
DemConfigSet
DemGeneral
DemCallbackDTCStatusChangeds
DemClients
DemDataClass
DemDidClass
DemEnableConditionGroups
DemEnableConditions
DemExtendedDataClass
DemExtendedDataRecordClass
DemFreezeFrameClass
DemFreezeFrameRecNumClass
DemFreezeFrameRecordClass
DemGroupOfDTCs
DemIndicators
DemMILGroups
DemNvRamBlockIds
DemOperat
DemRatioId
DemStorage
DemStorage
DemTimeSe
Create DemNvRamBlockId Container
Copy
Paste ARXML
Ctrl+Shift+V
Show Properties

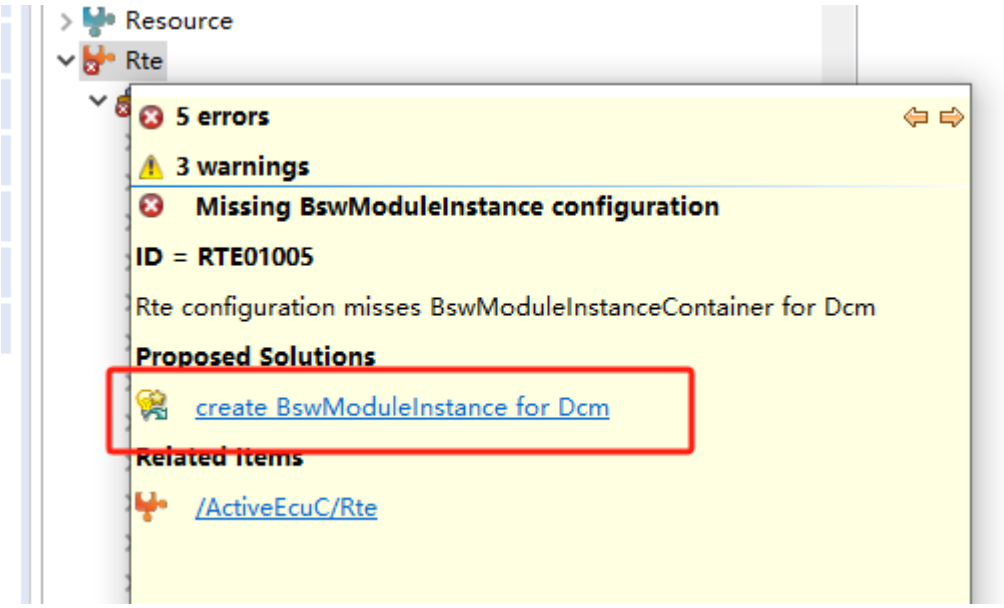

Basic Editor
Project Settings
Dem
DemGeneral
DemNvRamBlockIds
DemNvRamBlockAdmin, DemNvRamBlockStatus, DemNvRamBlockPrimary0, DemNvRamBlockPrimary1, DemNvRamBlockPrimary2, DemNvRamBlockPrimary3
DemConfigSet
DemGeneral
DemCallbackDTCStatusChangeds
DemClients
DemDataClass
DemDidClass
DemEnableConditionGroups
DemEnableConditions
DemExtendedDataClass
DemExtendedDataRecordClass
DemFreezeFrameClass
DemFreezeFrameRecNumClass
DemFreezeFrameRecordClass
DemGroupOfDTCs
DemIndicators
DemMILGroups
DemNvRamBlockIds
DemNvRamBlockAdmin
DemNvRamBlockPrimary0
DemNvRamBlockPrimary1
DemNvRamBlockPrimary2
DemNvRamBlockPrimary3
DemNvRamBlockPrimary4
DemNvRamBlockPrimary5
DemNvRamBlockPrimary6
DemNvRamBlockPrimary7
DemNvRamBlockStatus
DemNvRamBlockIds | NvRam BlockId Index | NvRam BlockId Ref | NvRam BlockId Type | User Defined Memory Ref
DemNvRamBlockAdmin | | NvMBlockDemAdmin | DEM_NVBLOCK_ADMIN
DemNvRamBlockPrimary0 | 0 | NvMBlockDemPrimary0 | DEM_NVBLOCK_PRIMARY
DemNvRamBlockPrimary1 | 1 | NvMBlockDemPrimary1 | DEM_NVBLOCK_PRIMARY
DemNvRamBlockPrimary2 | 2 | NvMBlockDemPrimary2 | DEM_NVBLOCK_PRIMARY
DemNvRamBlockPrimary3 | 3 | NvMBlockDemPrimary3 | DEM_NVBLOCK_PRIMARY
DemNvRamBlockPrimary4 | 4 | NvMBlockDemPrimary4 | DEM_NVBLOCK_PRIMARY
DemNvRamBlockPrimary5 | 5 | NvMBlockDemPrimary5 | DEM_NVBLOCK_PRIMARY
DemNvRamBlockPrimary6 | 6 | NvMBlockDemPrimary6 | DEM_NVBLOCK_PRIMARY
DemNvRamBlockPrimary7 | 7 | NvMBlockDemPrimary7 | DEM_NVBLOCK_PRIMARY
DemNvRamBlockStatus | | NvMBlockDemStatus | DEM_NVBLOCK_STATUS

同步swc

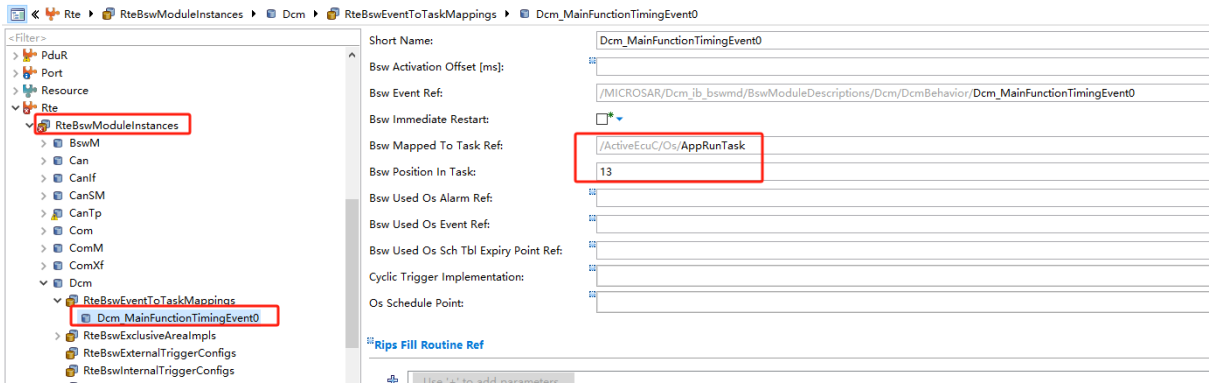


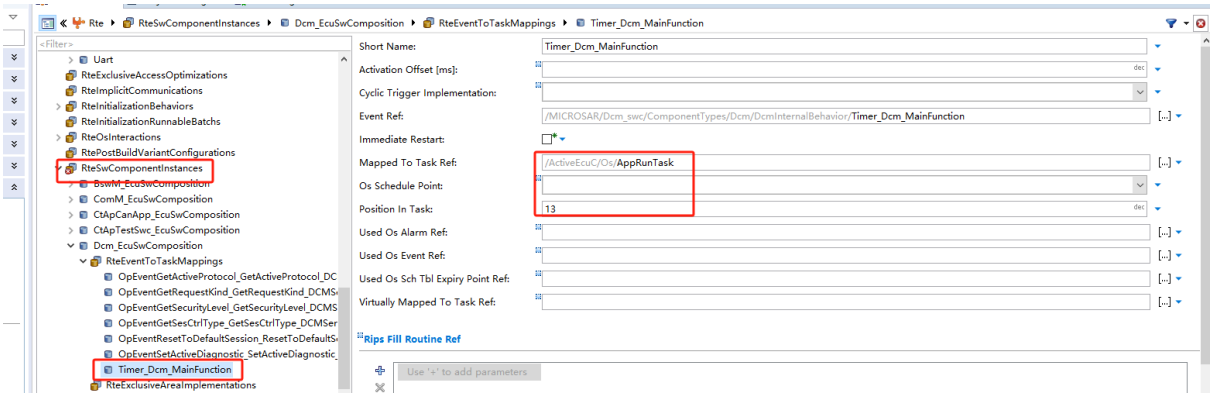
4.7、Rte消错

Rte主要是为新增的模块的mainfunction map到task

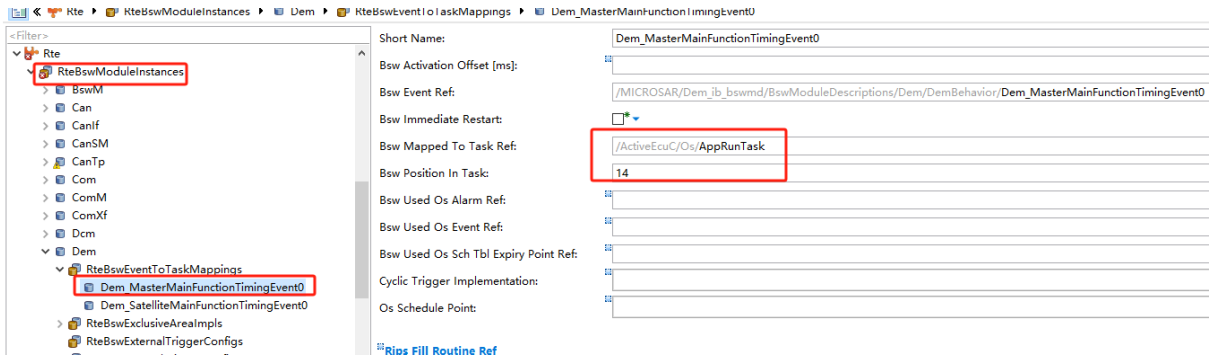


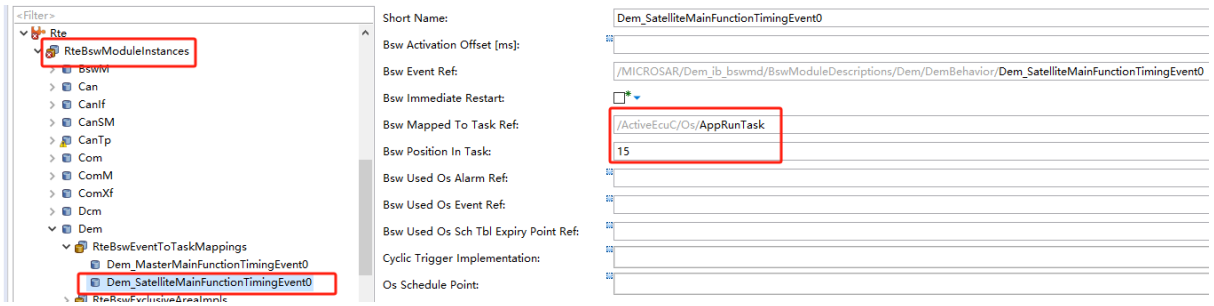
Dcm

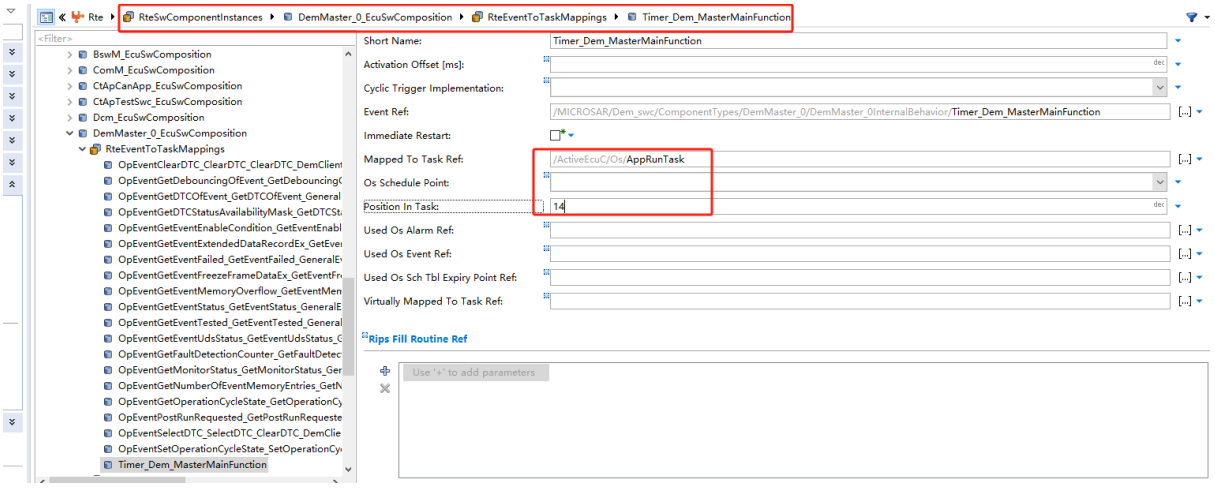

Rte ▸ RteBswModuleInstances ▸ Dcm ▸ RteBswEventToTaskMappings ▸ Dcm_MainFunctionTimingEvent0
Short Name: Dcm_MainFunctionTimingEvent0
Bsw Activation Offset [ms]:
Bsw Event Ref: /MICROSAR/Dcm_ib_bswmd/BswModuleDescriptions/Dcm/DcmBehavior/Dcm_MainFunctionTimingEvent0
Bsw Immediate Restart:
Bsw Mapped To Task Ref: /ActiveEcuC/Os/AppRunTask
Bsw Position In Task: 13
Bsw Used Os Alarm Ref:
Bsw Used Os Event Ref:
Bsw Used Os Sch Tbl Expiry Point Ref:
Cyclic Trigger Implementation:
Os Schedule Point:
Rips Fill Routine Ref

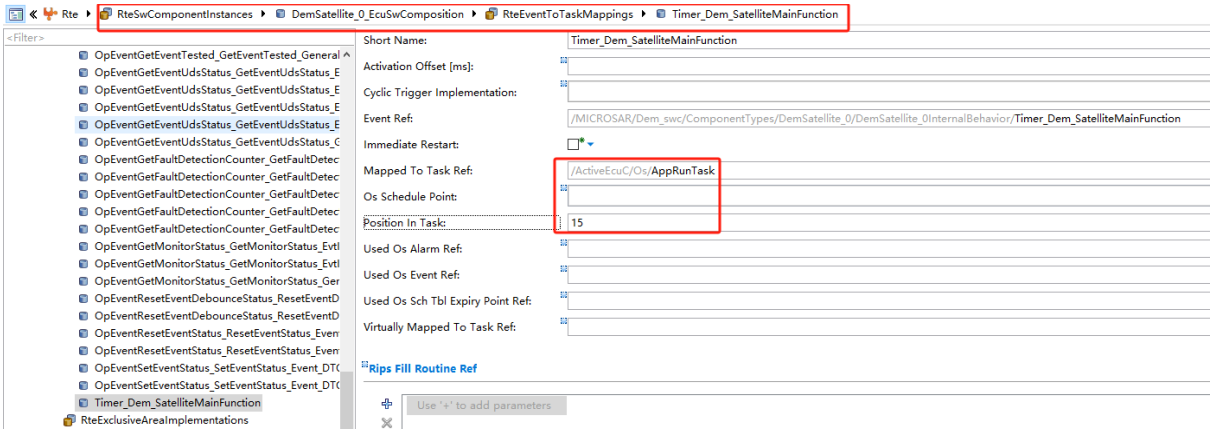

Rte ▸ RteSwComponentInstances ▸ Dcm_EcuSwComposition ▸ RteEventToTaskMappings ▸ Timer_Dcm_MainFunction
Short Name: Timer_Dcm_MainFunction
Activation Offset [ms]:
Cyclic Trigger Implementation:
Event Ref: /MICROSAR/Dcm_swc/ComponentTypes/Dcm/DcmInternalBehavior/Timer_Dcm_MainFunction
Immediate Restart:
Mapped To Task Ref: /ActiveEcuC/Os/AppRunTask
Os Schedule Point:
Position In Task: 13
Used Os Alarm Ref:
Used Os Event Ref:
Used Os Sch Tbl Expiry Point Ref:
Virtually Mapped To Task Ref:
Rips Fill Routine Ref
Use '+' to add parameters

Dem

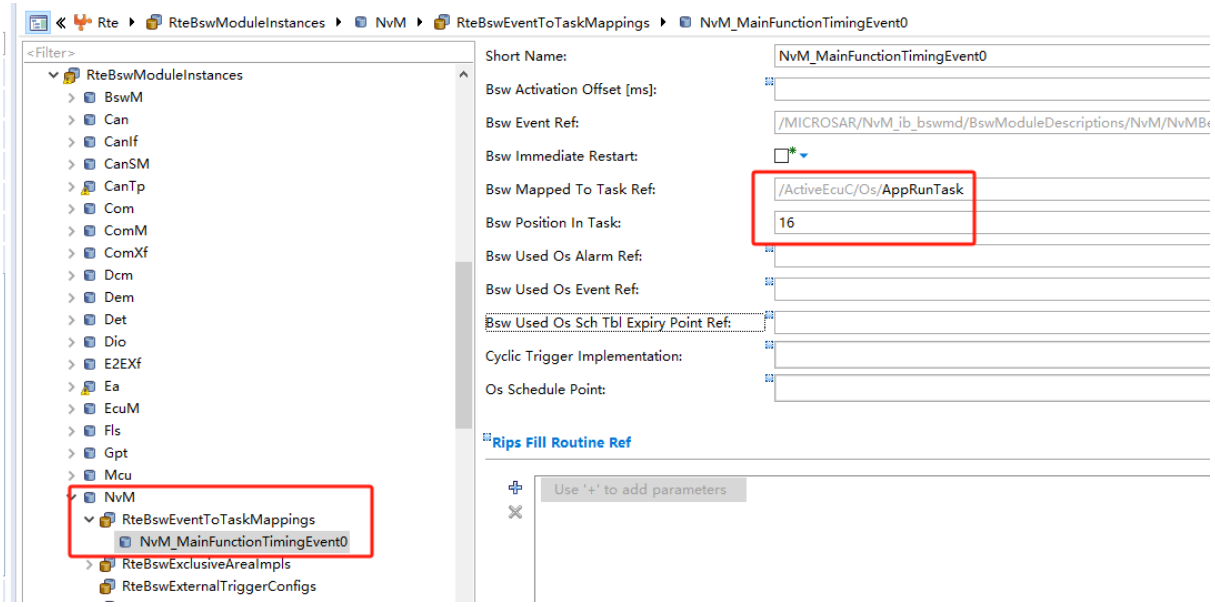

Short Name: Dem_MasterMainFunctionTimingEvent0
Bsw Activation Offset [ms]:
Bsw Event Ref: /MICROSAR/Dem_ib_bswmd/BswModuleDescriptions/Dem/DemBehavior/Dem_MasterMainFunctionTimingEvent0
Bsw Immediate Restart:
Bsw Mapped To Task Ref: /ActiveEcuC/Os/AppRunTask
Bsw Position In Task: 14
Bsw Used Os Alarm Ref:
Bsw Used Os Event Ref:
Bsw Used Os Sch Tbl Expiry Point Ref:
Cyclic Trigger Implementation:
Os Schedule Point:
Rips Fill Routine Ref

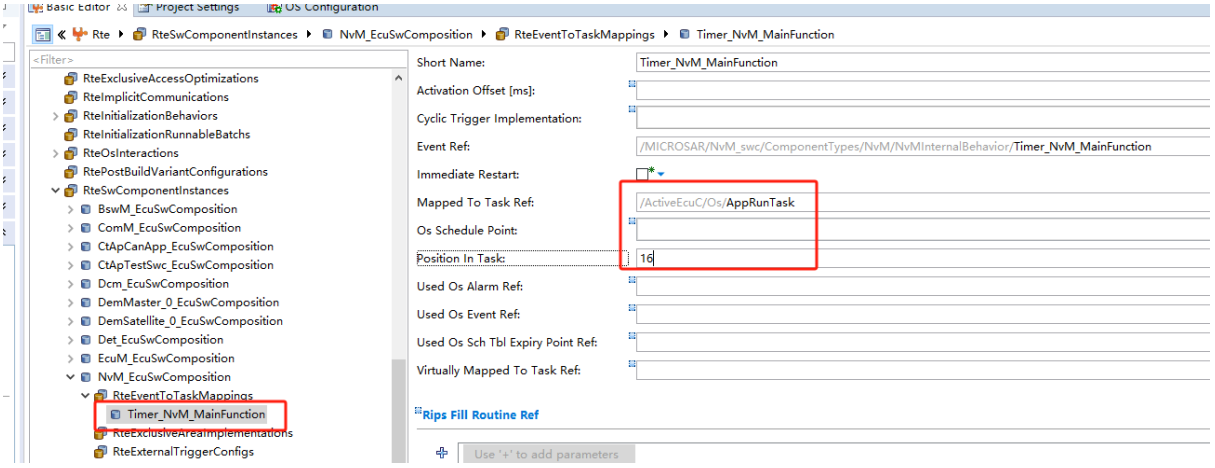

Short Name: Dem_SatelliteMainFunctionTimingEvent0
Bsw Activation Offset [ms]:
Bsw Event Ref: /MICROSAR/Dem_ib_bswmd/BswModuleDescriptions/Dem/DemBehavior/Dem_SatelliteMainFunctionTimingEvent0
Bsw Immediate Restart:
Bsw Mapped To Task Ref: /ActiveEcuC/Os/AppRunTask
Bsw Position In Task: 15
Bsw Used Os Alarm Ref:
Bsw Used Os Event Ref:
Bsw Used Os Sch Tbl Expiry Point Ref:
Cyclic Trigger Implementation:
Os Schedule Point:


Rte ▸ RteSwComponentInstances ▸ DemMaster_0_EcuSwComposition ▸ RteEventToTaskMappings ▸ Timer_Dem_MasterMainFunction
Short Name: Timer_Dem_MasterMainFunction
Activation Offset [ms]:
Cyclic Trigger Implementation:
Event Ref: /MICROSAR/Dem_swc/ComponentTypes/DemMaster_0/DemMaster_0InternalBehavior/Timer_Dem_MasterMainFunction
Immediate Restart:
Mapped To Task Ref: /ActiveEcuC/Os/AppRunTask
Os Schedule Point:
Position In Task: 14
Used Os Alarm Ref:
Used Os Event Ref:
Used Os Sch Tbl Expiry Point Ref:
Virtually Mapped To Task Ref:
Rips Fill Routine Ref
Use '+' to add parameters

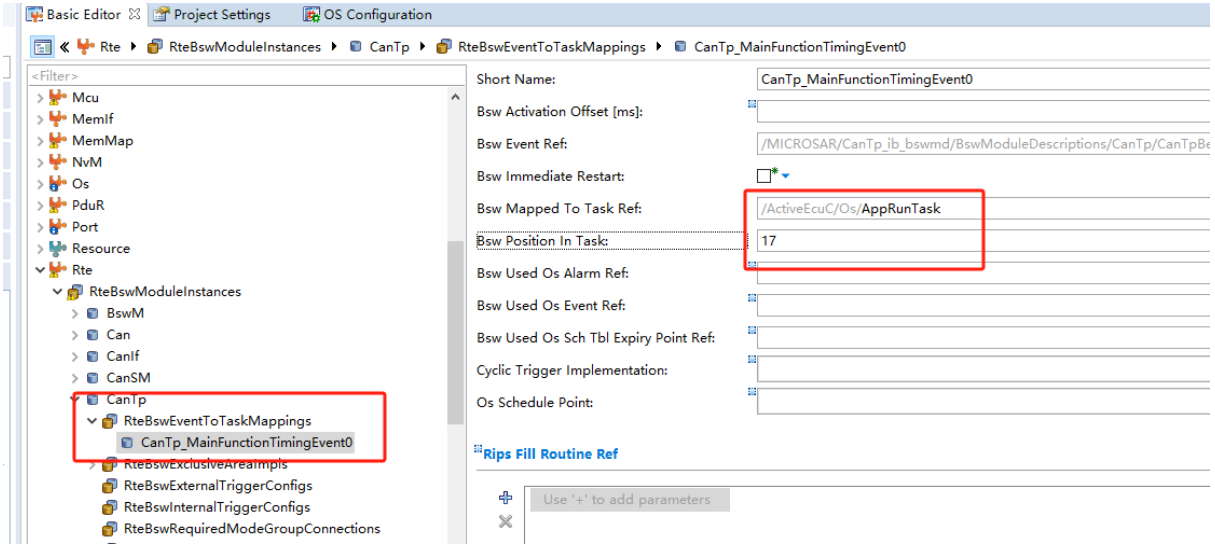

Rte ▸ RteSwComponentInstances ▸ DemSatellite_0_EcuSwComposition ▸ RteEventToTaskMappings ▸ Timer_Dem_SatelliteMainFunction
Short Name: Timer_Dem_SatelliteMainFunction
Activation Offset [ms]:
Cyclic Trigger Implementation:
Event Ref: /MICROSAR/Dem_swc/ComponentTypes/DemSatellite_0/DemSatellite_0InternalBehavior/Timer_Dem_SatelliteMainFunction
Immediate Restart:
Mapped To Task Ref: /ActiveEcuC/Os/AppRunTask
Os Schedule Point:
Position In Task: 15
Used Os Alarm Ref:
Used Os Event Ref:
Used Os Sch Tbl Expiry Point Ref:
Virtually Mapped To Task Ref:
Rips Fill Routine Ref
Use '+' to add parameters

NvM


Rte ▸ RteBswModuleInstances ▸ NvM ▸ RteBswEventToTaskMappings ▸ NvM_MainFunctionTimingEvent0
RteBswModuleInstances
BswM
Can
CanIf
CanSM
CanTp
Com
ComM
ComXf
Dcm
Dem
Det
Dio
E2EXf
Ea
EcuM
Fls
Gpt
Mcu
NvM
RteBswEventToTaskMappings
NvM_MainFunctionTimingEvent0
RteBswExclusiveAreaImpls
RteBswExternalTriggerConfigs
Short Name: NvM_MainFunctionTimingEvent0
Bsw Activation Offset [ms]:
Bsw Event Ref:
Bsw Immediate Restart:
Bsw Mapped To Task Ref: /ActiveEcuC/Os/AppRunTask
Bsw Position In Task: 16
Bsw Used Os Alarm Ref:
Bsw Used Os Event Ref:
Bsw Used Os Sch Tbl Expiry Point Ref:
Cyclic Trigger Implementation:
Os Schedule Point:
Rips Fill Routine Ref
Use '+' to add parameters

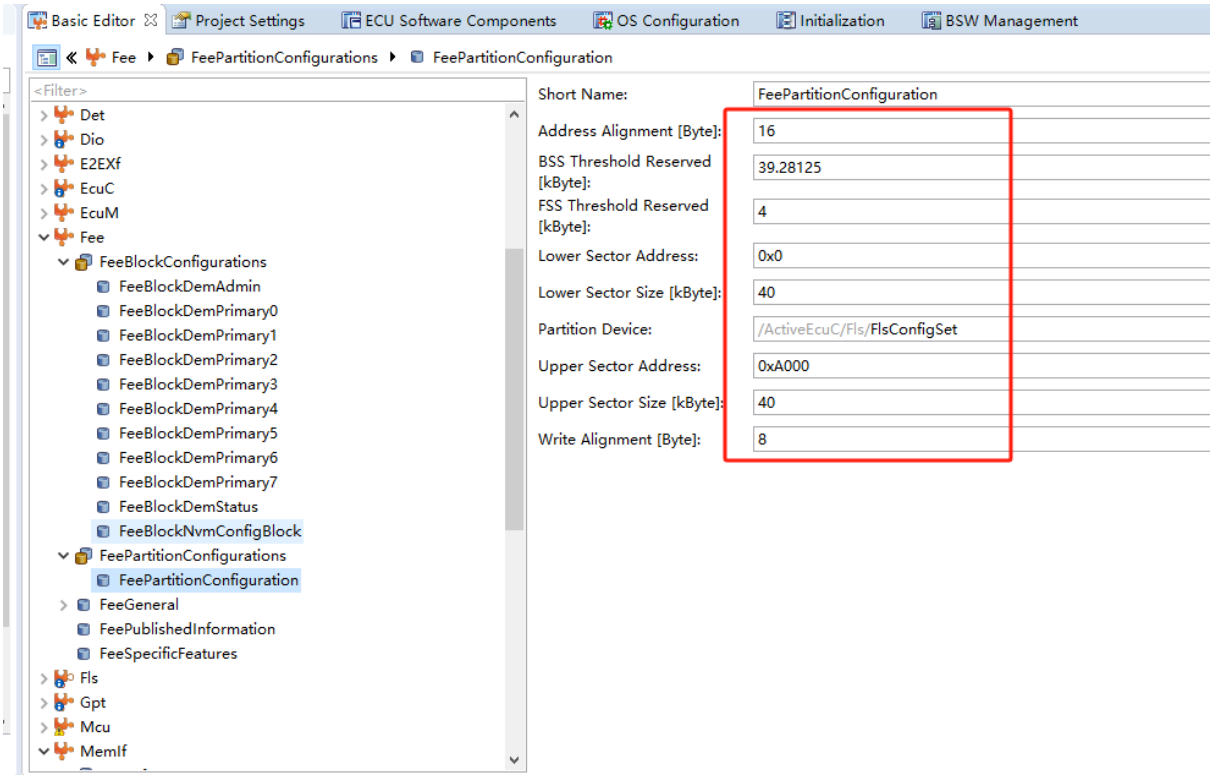


CanTp

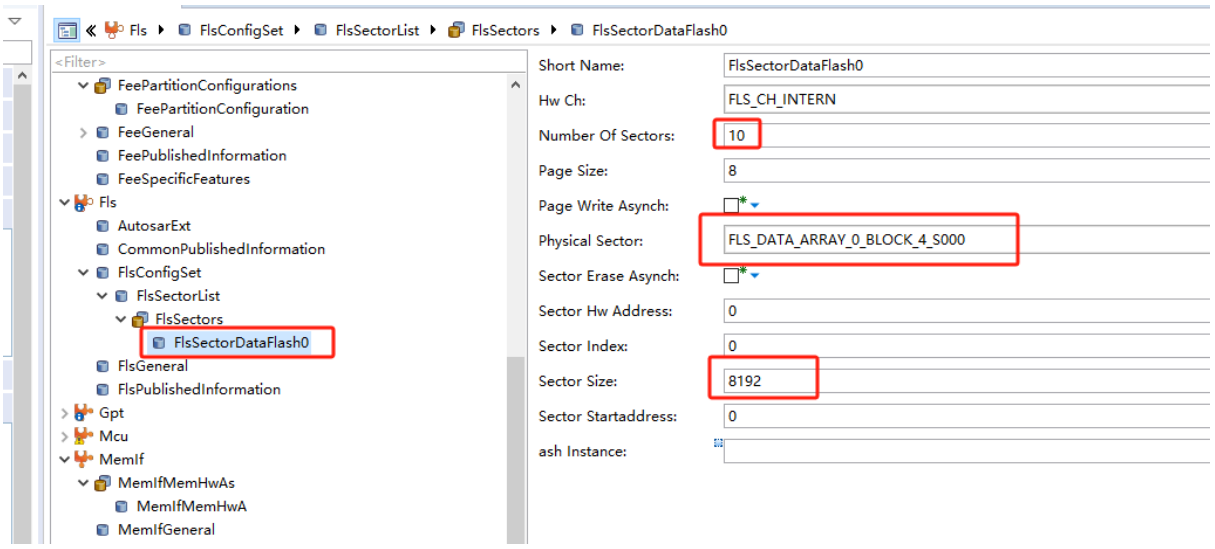


4.8、Fee消错，Fls修改

Fee 我们配置用Fls 的80K

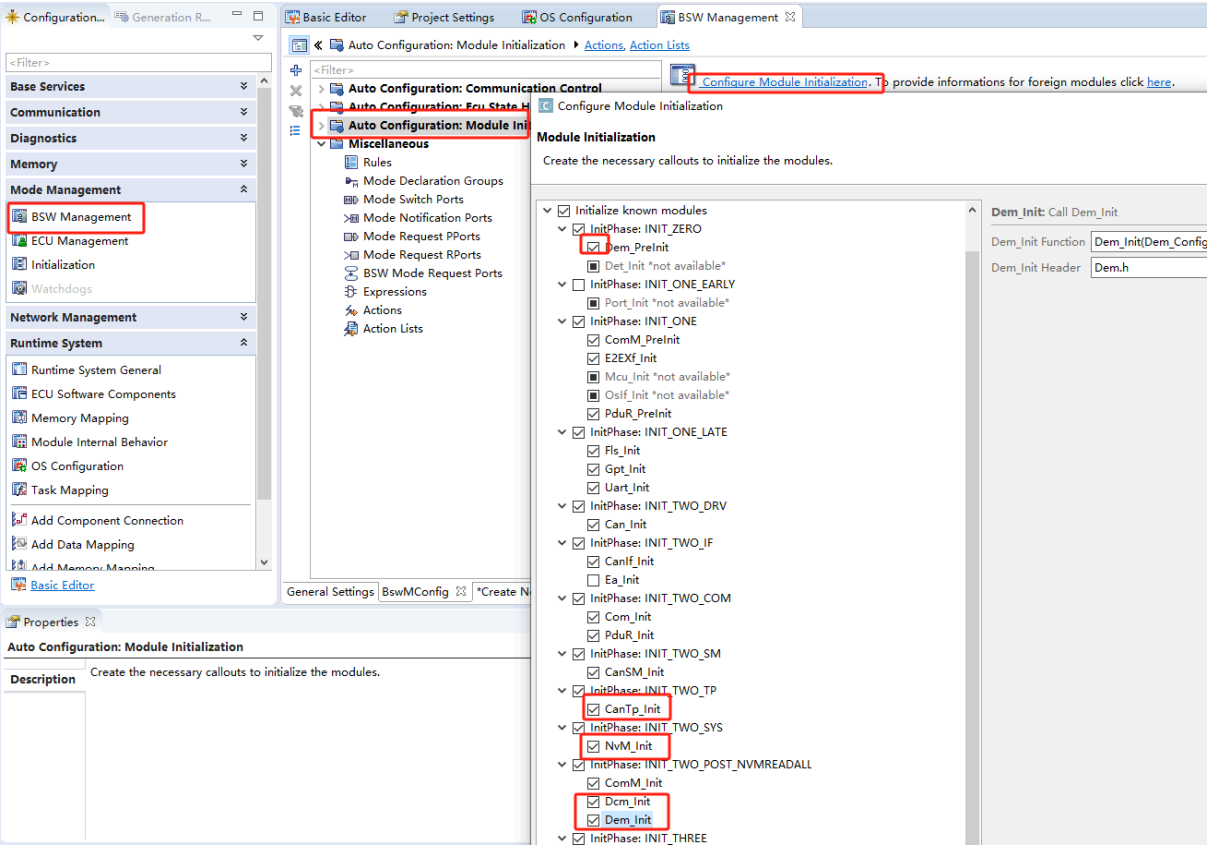


Fls我们也修改只保留一个Sectors，选择从第一个sector开始，使用10个， 一共就是80K

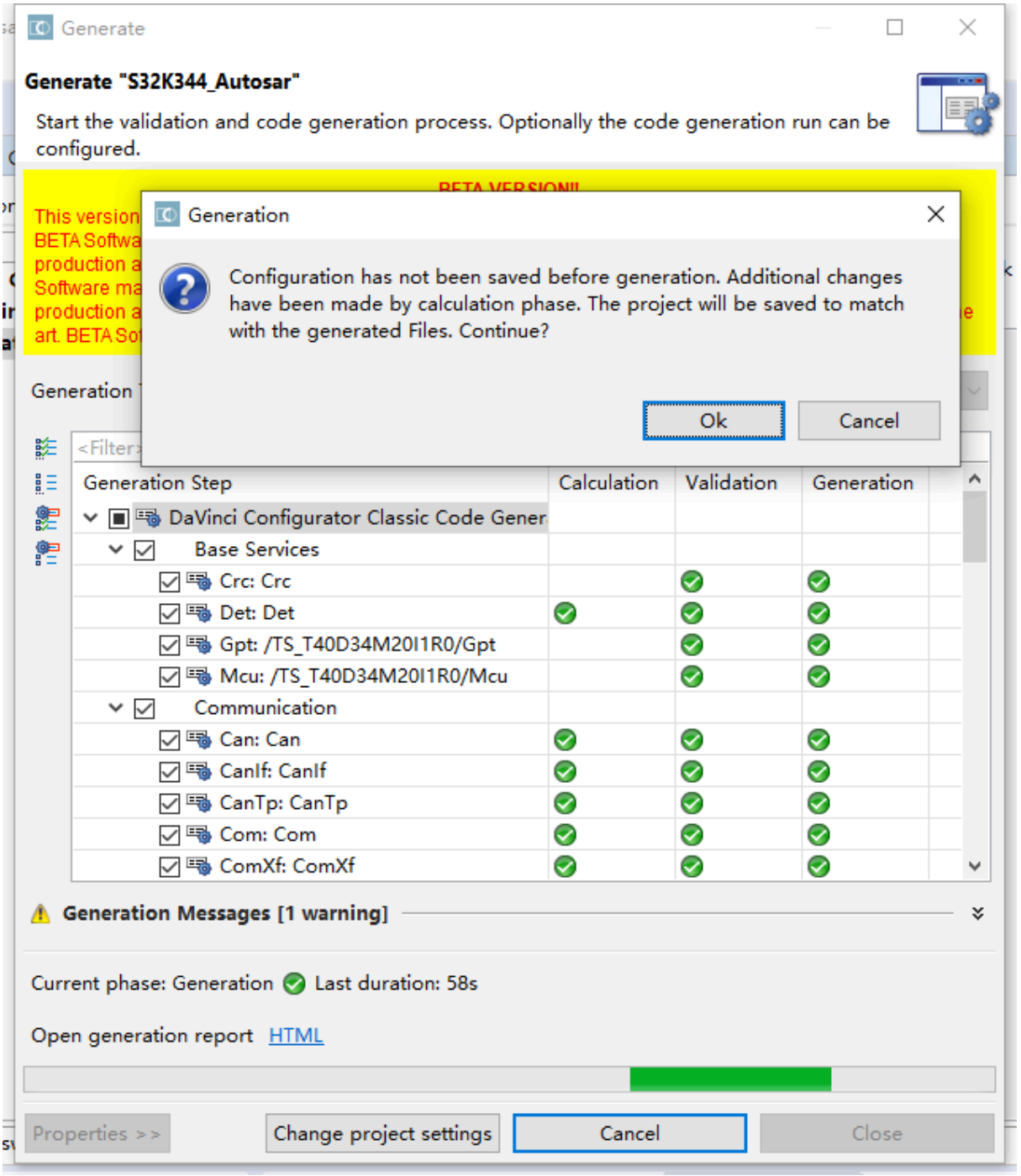


4.8、BswM添加Dem\CanTp\Nvm\Dcm\Fee的init，不添加Ea的

- [ ] Execute NvM ReadAll
- [ ] Enable Interrupts
- [x] Initialize known modules
  - [x] InitPhase: INIT_ZERO
    - [x] Dem_PreInit
    - [■] Det_Init *not available*
  - [ ] InitPhase: INIT_ONE_EARLY
    - [■] Port_Init *not available*
  - [x] InitPhase: INIT_ONE
    - [x] ComM_PreInit
    - [x] E2EXf_Init
    - [■] Mcu_Init *not available*
    - [■] OsIf_Init *not available*
    - [x] PduR_PreInit
  - [x] InitPhase: INIT_ONE_LATE
    - [x] Fls_Init
    - [x] Gpt_Init
    - [x] Uart_Init
  - [x] InitPhase: INIT_TWO_DRV
    - [x] Can_Init
  - [x] InitPhase: INIT_TWO_IF
    - [x] CanIf_Init
    - [x] Fee_Init
  - [x] InitPhase: INIT_TWO_COM
    - [x] Com_Init
    - [x] PduR_Init
  - [x] InitPhase: INIT_TWO_SM
    - [x] CanSM_Init
  - [x] InitPhase: INIT_TWO_TP
    - [x] CanTp_Init
  - [x] InitPhase: INIT_TWO_SYS
    - [x] NvM_Init
  - [x] InitPhase: INIT_TWO_POST_NVMREADALL
    - [x] ComM_Init
    - [x] Dcm_Init
    - [x] Dem_Init
  - [x] InitPhase: INIT_THREE
    - [x] ComM_PostInit
    - [x] Rte_Start



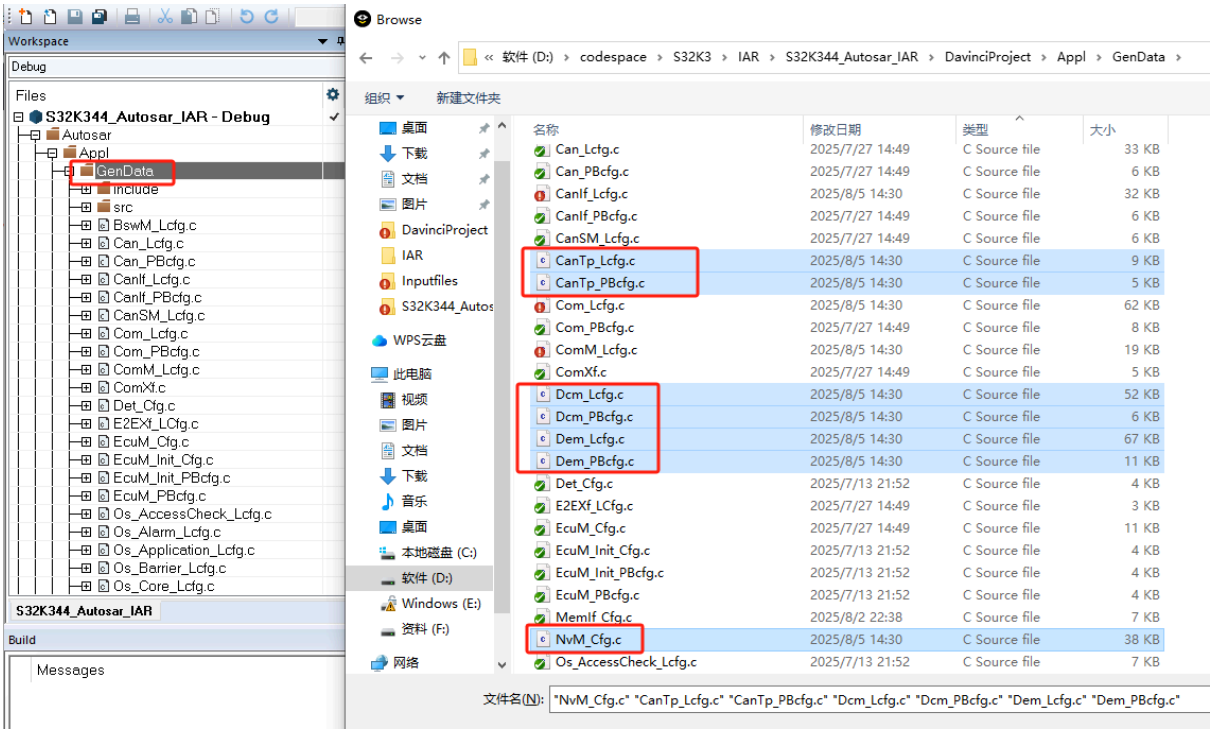
4.9、生成所以模块　除Ea的模块

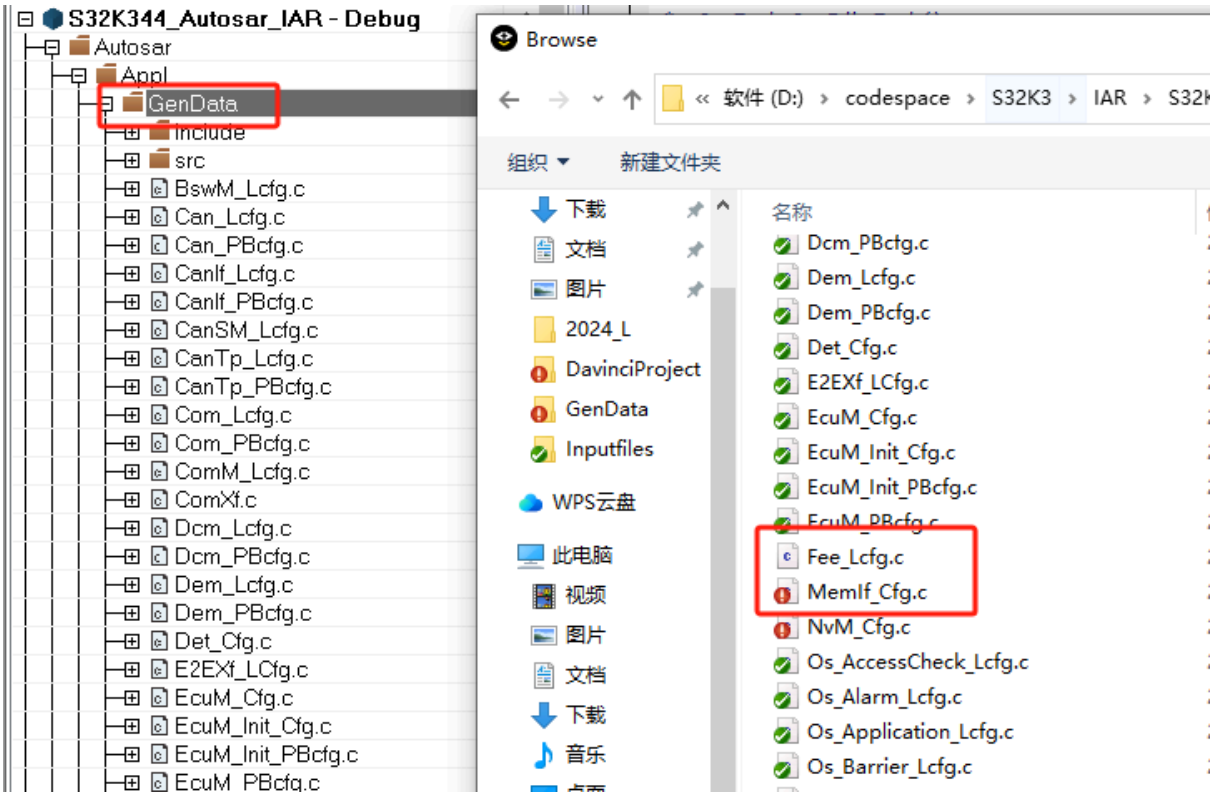


五、IAR添加对应文件编译

5.1、添加davinci生成的模块文件到GenData

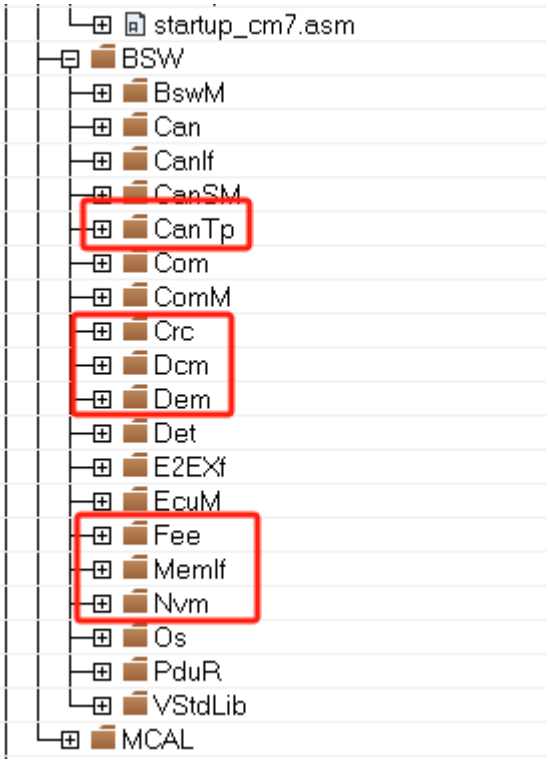
GenData下添加davinci生成的CanTp、Dcm、Dem、Nvm、MemIf、Fee的文件

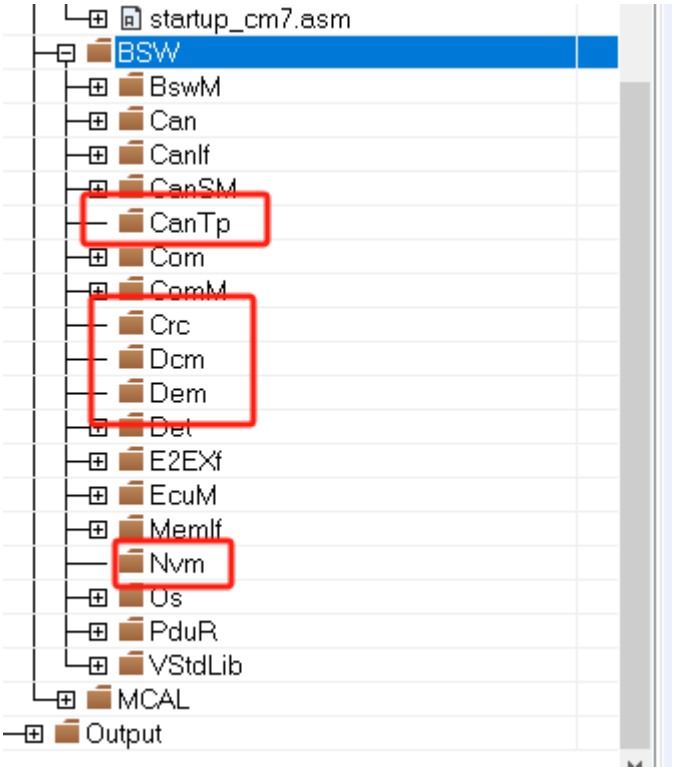




5.2、添加新增的Bsw模块的SIP包静态代码

添加CanTp\Crc\Dcm\Dem\Nvm\Fee的静态代码





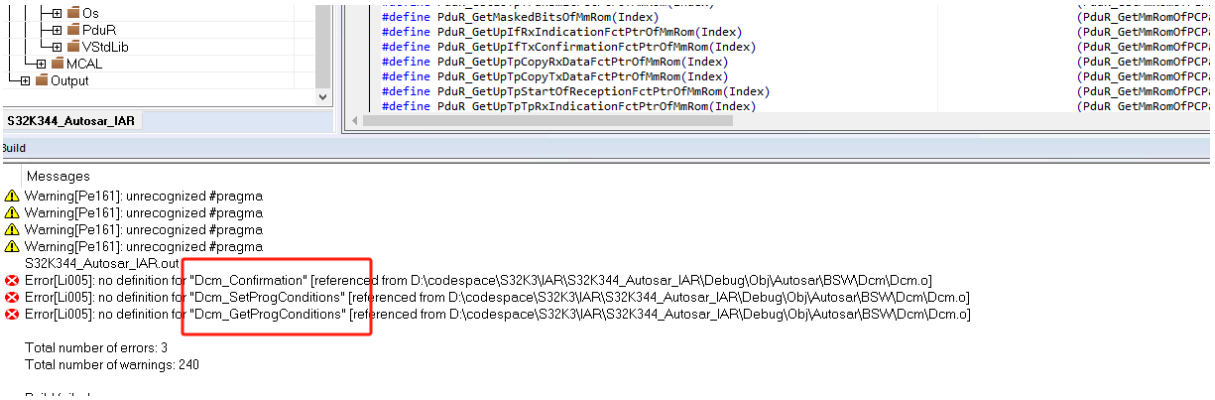
并且添加以下头文件路径到工程

$PROJ_DIR$\CBD2300384_D00\Components\CanTp\Implementation
$PROJ_DIR$\CBD2300384_D00\Components\Nvm\Implementation
$PROJ_DIR$\CBD2300384_D00\Components\Dcm\Implementation

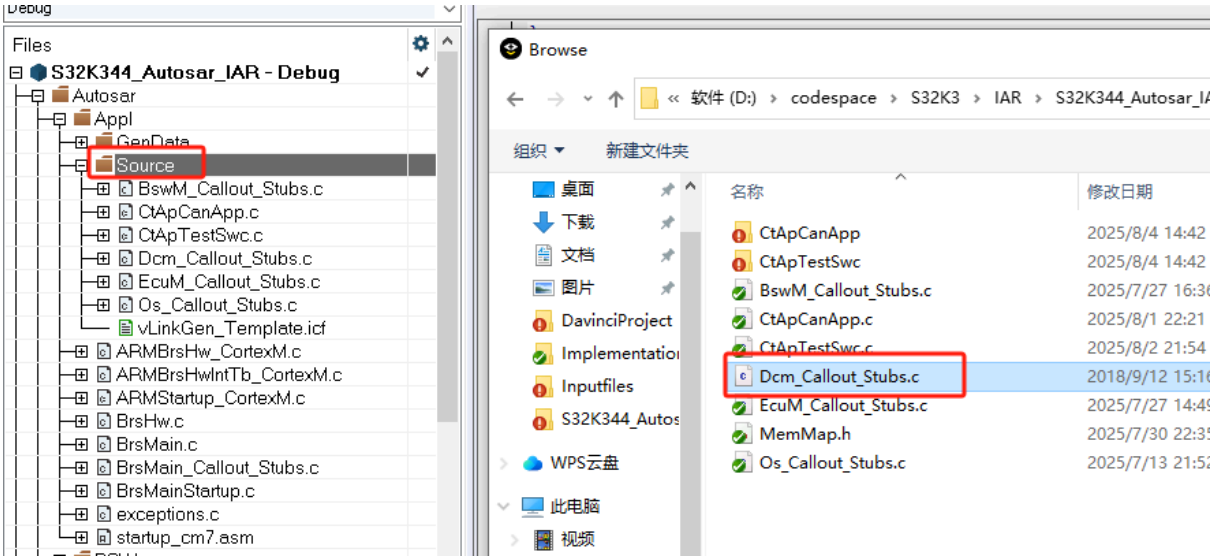
$PROJ_DIR$\CBD2300384_D00\Components\Dem\Implementation
$PROJ_DIR$\CBD2300384_D00\Components\Crc\Implementation

$PROJ_DIR$\CBD2300384_D00\Components\Fee\Implementation

5.3、缺少Dcm_Callout_Stubs.c报错



由于SIP包没有Dcm_Callout_Stubs.c ，我们从别的SIP包添加过来即可，基本都是通用，添加到GenData\Source下



5.4、MemIf报错

```
Messages

MemIf.c
Warning[Pa181]: incompatible redefinition of macro "INLINE" (defined at line 114)
Warning[Pa181]: incompatible redefinition of macro "LOCAL_INLINE" (defined at line 120)
Warning[Pa181]: incompatible redefinition of macro "INTERRUPT_FUNC" (defined at line 125)
S32K344_Autosar_IAR.out
Error[Li005]: no definition for "MemIf_MemHwaApis" [referenced from D:\codespace\S32K3\IAR\S32K344_Autosar_IAR\Debug\Obj\Autosar\BSW\MemIf\MemIf.o]
Error[Li005]: no definition for "MemIf_NumberOfDevices" [referenced from D:\codespace\S32K3\IAR\S32K344_Autosar_IAR\Debug\Obj\Autosar\BSW\MemIf\MemIf.o]

Total number of errors: 2
Total number of warnings: 3
```

由于我们没有生成Ea导致的，这里先把MemIf.c 里面涉及这两个函数都注释先

```
/*****************************************************************************
 *  INCLUDES
 *****************************************************************************/
#include "Std_Types.h"
#include "MemIf.h"
/*****************************************************************************
 *  VERSION CHECK
 *****************************************************************************/
```

```
/*****************************************************************************
 *  INCLUDES
 *****************************************************************************/
#include "Std_Types.h"
#include "MemIf.h"
#define MemIf_NumberOfDevices 1
/*****************************************************************************
 *  VERSION CHECK
 *****************************************************************************/
```

```
C:\Users\Administrator\AppData\Local\Temp\TortoiseGit\MemIf-1e2d8dd8.000.c
2025/8/5 15:07:23  22,808 字节  只读  C,C++,C#,ObjC 源代码  ANSI  PC

    retVal = MemIf_MemHwaApis[DeviceIndex].Read(BlockNumber, BlockOffset, DataBufferPtr, Length); /* SBSW_MEMIF_02 */
    retVal = MemIf_MemHwaApis[DeviceIndex].Write(BlockNumber, DataBufferPtr); /* SBSW_MEMIF_02 */
    retVal = MemIf_MemHwaApis[DeviceIndex].InvalidateBlock(BlockNumber); /* SBSW_MEMIF_02 */
    retVal = MemIf_MemHwaApis[DeviceIndex].EraseImmediateBlock(BlockNumber); /* SBSW_MEMIF_02 */
    MemIf_MemHwaApis[DeviceIndex].Cancel(); /* SBSW_MEMIF_02 */
      MemIf_StatusMask |= (uint8) ((uint8) 1u << (uint8) MemIf_MemHwaApis[MemIf_LoopCounter].GetStatus()); /* SBSW_MEMIF_02
     retVal = MemIf_MemHwaApis[DeviceIndex].GetStatus(); /* SBSW_MEMIF_02 */
    retVal = MemIf_MemHwaApis[DeviceIndex].GetJobResult(); /* SBSW_MEMIF_02 */
    MemIf_MemHwaApis[MemIf_LoopCounter].SetMode(Mode); /* SBSW_MEMIF_02 */
```

```
D:\codespace\S32K3\IAR\S32K344_Autosar_IAR\CBD2300384_D00\Components\MemIf\Implementation\MemIf.c
2025/8/4 15:02:34  22,872 字节  C,C++,C#,ObjC 源代码  ANSI  PC

#define MemIf_NumberOfDevices 1
    retVal = 0;//MemIf_MemHwaApis[DeviceIndex].Read(BlockNumber, BlockOffset, DataBufferPtr, Length); /* SBSW_MEMIF_02
    retVal = 0;//MemIf_MemHwaApis[DeviceIndex].Write(BlockNumber, DataBufferPtr); /* SBSW_MEMIF_02 */
    retVal = 0;//MemIf_MemHwaApis[DeviceIndex].InvalidateBlock(BlockNumber); /* SBSW_MEMIF_02 */
    retVal = 0;//MemIf_MemHwaApis[DeviceIndex].EraseImmediateBlock(BlockNumber); /* SBSW_MEMIF_02 */
    //MemIf_MemHwaApis[DeviceIndex].Cancel(); /* SBSW_MEMIF_02 */
      MemIf_StatusMask |= (uint8) ((uint8) 1u << (uint8) 0/*MemIf_MemHwaApis[MemIf_LoopCounter].GetStatus()*/); /*
     retVal = 0;//MemIf_MemHwaApis[DeviceIndex].GetStatus(); /* SBSW_MEMIF_02 */
    retVal = 0;//MemIf_MemHwaApis[DeviceIndex].GetJobResult(); /* SBSW_MEMIF_02 */
    //MemIf_MemHwaApis[MemIf_LoopCounter].SetMode(Mode); /* SBSW_MEMIF_02 */
```

至此，编译都完成且没有报错了

六、运行

手动发一条 1001， 回复的是正常的5001

行	发送	触发	报文名称	标识符	通道	类型	DLC	BRS	D0	D1	D2	D3	D4	D5	D6	D7	注释
1	▶	手动	NewMsg	700	1	标准数据帧	8	☐	02	10	01	00	00	00	00	00	

字节索引	字节 0	字节 1	字节 2	字节 3	字节 4	字节 5	字节 6	字节 7
0	02	10	01	00	00	00	00	00

绝对时间	计数	...	标...	帧率	报...	类型	方向	DLC	数据长度	BRS	ESI	00	01	02	03	04
624.303070	68667	1	200	10		数据帧	接收	8	8	-	-	55	F3	C2	00	00
624.373071	68675	1	300	100		数据帧	接收	8	8	-	-	C9	00	00	00	00
622.703570	68491	1	700	0		数据帧	发送	8	8	-	-	02	10	01	00	00
622.713096	68493	1	708	0		数据帧	接收	8	8	-	-	06	50	01	00	32